

A 2-D Unstructured Compressible Navier–Stokes Solver: Method, Verification, and Benchmark Results

ufvm2d — submitted solver

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Abstract

This report documents `ufvm2d`, an original cell-centred finite-volume solver for the two-dimensional compressible Euler and laminar Navier–Stokes equations on unstructured mixed triangular/quadrilateral meshes. The spatial discretisation is second order through least-squares gradients with Venkatakrishnan limiting; inviscid fluxes use a Roe scheme with a Harten–Yee wave-speed floor; viscous fluxes use Newtonian stress with Fourier heat conduction and directionally corrected face gradients. Steady cases are marched with a backward-Euler pseudo-time scheme solved by symmetric Gauss–Seidel relaxation about an analytic 4×4 flux Jacobian; the unsteady cylinder case uses a true dual-time BDF2 method with an outer physical-time loop and frozen histories during inner iterations. Parallelism is distributed-memory MPI with METIS k -way partitioning: the global mesh is read and partitioned once on rank 0 and then released, so no rank holds the full mesh or the full conservative state during iterations, and halo data moves only through neighbour-scoped non-blocking exchanges. Verification evidence includes freestream preservation, discrete conservation, gradient exactness on linear fields, and an analytic-versus-finite-difference Jacobian comparison. All required cases were run to completion; the convergence status, force coefficients, and known limitations of each are reported without adjustment.

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1 Introduction

The objective of this benchmark is a working, original, parallel finite-volume solver for the two-dimensional compressible Navier–Stokes equations, exercised on a fixed set of aerofoil and cylinder cases and documented to the standard of a technical CFD report. Nothing in the solver core is adapted from an existing CFD code: the flux functions, gradient and limiter machinery, boundary treatments, implicit relaxation, mesh welding, and partitioning glue were all written from the governing mathematics for this submission. Infrastructure libraries (MPI, CGNS, HDF5, METIS, `nlohmann/json`) are used as intended.

The required cases are seven steady runs and one unsteady run:

- NACA0012 at $\alpha = 0^\circ$, inviscid, $M_\infty \in \{0.15, 0.80, 2.00\}$;
- NACA0012 at $\alpha = 0^\circ$, laminar $Re = 5000$, $M_\infty \in \{0.15, 0.80, 2.00\}$;
- circular cylinder, laminar $Re = 20$, $M_\infty = 0.10$ (steady);
- circular cylinder, laminar $Re = 200$, $M_\infty = 0.10$ (unsteady vortex shedding).

The report is organised to follow the physics and the code: governing equations and nondimensionalisation (Section 2), mesh handling (Section 3), spatial discretisation (Section 4), boundary conditions (Section 5), time integration (Section 6), parallelisation (Section 7), reproducibility and automated checks (Section 8), verification (Section 9), results (Section 10), parallel validation (Section 11), and an honest account of what remains wrong or unproven (Section 14). Every quantitative statement is taken from a file in the submitted `results/` tree; where a case did not meet its nominal target, that is stated in the text rather than absorbed into a summary.

2 Governing Equations and Nondimensionalisation

2.1 Conservation form

The solver advances the conservative state

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T, \quad (1)$$

governed by the two-dimensional compressible Navier–Stokes equations

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \quad (2)$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \quad (3)$$

The gas is calorically perfect, so

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}, \quad (4)$$

with $\gamma = 1.4$ and $R = 1$ for every case. All equation-of-state knowledge is confined to a single **PerfectGas** class, which the flux routines take by reference; a different closure is a new class rather than an edit to the flux code.

2.2 Viscous fluxes

The viscous fluxes follow from the Newtonian stress tensor with the Stokes hypothesis ($\lambda = -\frac{2}{3}\mu$),

$$\tau_{xx} = 2\mu \frac{\partial u}{\partial x} - \frac{2}{3}\mu \nabla \cdot \mathbf{u}, \quad \tau_{yy} = 2\mu \frac{\partial v}{\partial y} - \frac{2}{3}\mu \nabla \cdot \mathbf{u}, \quad \tau_{xy} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \quad (5)$$

together with Fourier heat conduction $\mathbf{q} = -k\nabla T$ and $k = \mu c_p / \text{Pr}$ at $\text{Pr} = 0.72$. The viscous flux normal to a face with unit normal $\mathbf{n} = (n_x, n_y)$ is

$$\mathbf{F}^v \cdot \mathbf{n} = \begin{bmatrix} 0 \\ \tau_{xx}n_x + \tau_{xy}n_y \\ \tau_{xy}n_x + \tau_{yy}n_y \\ (u\tau_{xx} + v\tau_{xy})n_x + (u\tau_{xy} + v\tau_{yy})n_y - \mathbf{q} \cdot \mathbf{n} \end{bmatrix}. \quad (6)$$

2.3 Nondimensionalisation and force coefficients

The case files supply a nondimensional freestream directly: $\rho_\infty = 1$, $|V_\infty| = 1$, and a pressure fixed by the target Mach number through $p_\infty = \rho_\infty a_\infty^2 / \gamma$ with $a_\infty = |V_\infty| / M_\infty$. For $M_\infty = 0.15$ this gives $p_\infty = 31.746$, for $M_\infty = 0.80$, $p_\infty = 1.1161$, for $M_\infty = 2.00$, $p_\infty = 0.17857$, and for the cylinder at $M_\infty = 0.10$, $p_\infty = 71.429$. The solver checks this redundancy on input and warns if the stated and implied pressures disagree by more than a relative 10^{-6} , but treats the stated pressure as authoritative; no warning was triggered by any supplied case.

The freestream velocity vector is $\mathbf{V}_\infty = |V_\infty|(\cos \alpha, \sin \alpha)$ with α the angle of attack in degrees ($\alpha = 0$ throughout), and the laminar viscosity follows from the case Reynolds number,

$$\mu = \frac{\rho_\infty |V_\infty| L_{\text{ref}}}{Re}, \quad (7)$$

which for $L_{\text{ref}} = 1$ gives $\mu = 2 \times 10^{-4}$ at $Re = 5000$, $\mu = 0.05$ at $Re = 20$, and $\mu = 5 \times 10^{-3}$ at $Re = 200$.

Wall loads are integrated from the pressure and the tangential viscous traction separately and normalised by the dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty |V_\infty|^2 = 0.5$ together with the case **reference.area** A_{ref} and **reference.length** L_{ref} , both unity for every supplied case:

$$C_D = \frac{F_x}{q_\infty A_{\text{ref}}}, \quad C_L = \frac{F_y}{q_\infty A_{\text{ref}}}, \quad C_m = \frac{M_z}{q_\infty A_{\text{ref}} L_{\text{ref}}}, \quad (8)$$

with the moment taken about the case **moment.center** ($x/c = 0.25$ for the aerofoil, the origin for the cylinder). Because $L_{\text{ref}} = 1$ equals the cylinder diameter, cylinder drag coefficients in this report are diameter-referenced and directly comparable with the standard literature values.

One consistency point matters more than it looks. The wall state entering Eq. (8) is the state reconstructed to the face centre — the same state the residual uses for the boundary flux (Section 5) — not the adjacent cell average. Using the cell average is the easier thing to write and is wrong in a specific way: the scheme driving the solution would be second order while the coefficients reported from it were first order, and the affected quantities are exactly C_L , C_D and the surface C_p that this report is about. An earlier version of this solver had that inconsistency. Correcting it moves the $Re = 20$ cylinder drag from 2.0171 to 2.0185, which is a 0.07% change and still inside the literature range — small, but the kind of small error that has no business being in the headline number.

3 Meshes, Boundary Tags, and Case Inputs

3.1 CGNS import

Both supplied meshes are HDF5-flavoured CGNS files with unstructured zones, `float64` coordinates, 1-based `int64` connectivity, and triangles and quadrilaterals in separate `Elements_t` sections (no `MIXED` sections). Boundary patches are `BAR_2` element sections whose element range is referenced from a `BC_t` node. An important practical detail: the base-level `Family_t` nodes carry `FamilyBC_t = "Null"`, so the mesh file does *not* state which family is a wall and which is a farfield. The physical type therefore comes from the `boundary_conditions` map in the case JSON, and a family present in the mesh but absent from that map is a hard input error rather than a silent default.

The reader ignores `CoordinateZ` when the base cell dimension is 2 (NACA0012.H2 stores a numerically zero z array, maximum magnitude 3.5×10^{-6}). After import the solver derives faces from cell connectivity, matches each boundary edge to exactly one cell, and builds the cell-to-cell adjacency graph; an edge that matches zero or more than two cells aborts the run as non-conforming input.

3.2 Two-zone welding for the cylinder

NACA0012.H2.cgns is a single zone and needs no special handling. CylinderB1.cgns contains *two* zones that must be welded into one conforming mesh: `dom-1`, an inner rectangle $[-2, 10] \times [-2, 2]$ with 7079 vertices and 6901 cells, and `dom-2`, the annular remainder out to radius 200 with 3236 vertices and 3284 cells.

Each zone carries a `ZoneGridConnectivity_t` with five `GridConnectivity_t` nodes (`con-2` through `con-6`) of type `Abutting1to1`, holding matched vertex point lists. A trap worth recording: `cg_n1to1` returns zero on these files, so the convenient one-to-one connectivity API finds nothing and a solver relying on it would treat the two zones as disconnected. The general connectivity API (`cg_nconns` followed by `cg_conn_read`) does return the five patches, and the solver uses that path. Welding merges 320 vertex pairs whose coordinates are exactly coincident, yielding a single mesh with 9995 global vertices and 10185 cells (500 triangles, 9685 quadrilaterals) and 20180 faces.

Geometric confirmation that the welded mesh is the intended geometry: the `WALL` family has 100 vertices all at radius 0.5 ± 10^{-16} about a centroid at $(1.3 \times 10^{-17}, 3.0 \times 10^{-17})$, i.e. the unit-diameter cylinder consistent with $L_{\text{ref}} = 1$, and the `FAR` family has 20 vertices at radius 200.

Table 1: Supplied meshes after import and (for the cylinder) zone welding. Counts are measured by the solver at run time and cross-checked against a standalone `h5py` inspection (`tools/inspect_mesh.py`).

Mesh	Zones	Vertices	Cells (tri + quad)	Faces	Wall / farfield edges
NACA0012.H2	1	15682	20816 (10752 + 10064)	36498	404 / 80
CylinderB1	2 welded	9995	10185 (500 + 9685)	20180	100 / 20

3.3 Geometry and boundary families

Cell centroids and areas are computed by the shoelace formula over each cell polygon; a negative signed area indicates a clockwise cell and is corrected so all volumes are positive. Face normals are the edge normals rotated to point from the left cell to the right cell, normalised to unit length, with the face area equal to the edge length in 2-D. On a boundary face the normal points out of

the domain, and the left cell is always an owned cell so that owned-face loops need no ownership test.

Table 2 lists the family-to-condition mapping and the run parameters for all eight cases exactly as they appear in the case files.

Table 2: Case inputs as supplied. Boundary families map to physical conditions through the case JSON; **bc-2/FAR** are farfield and **bc-4/WALL** are walls (slip for inviscid, no-slip adiabatic for laminar). p_∞ follows from M_∞ as described in Section 2.

Case	Mesh	M_∞	Re	p_∞	μ	Wall / run type
naca0012_m015_inviscid	NACA	0.15	—	31.746	0	slip / steady
naca0012_m080_inviscid	NACA	0.80	—	1.1161	0	slip / steady
naca0012_m200_inviscid	NACA	2.00	—	0.17857	0	slip / steady
naca0012_m015_laminar_re5000	NACA	0.15	5000	31.746	2×10^{-4}	no-slip / steady
naca0012_m080_laminar_re5000	NACA	0.80	5000	1.1161	2×10^{-4}	no-slip / steady
naca0012_m200_laminar_re5000	NACA	2.00	5000	0.17857	2×10^{-4}	no-slip / steady
cylinder_m010_laminar_re20	Cylinder	0.10	20	71.429	0.05	no-slip / steady
cylinder_m010_laminar_re200	Cylinder	0.10	200	71.429	5×10^{-3}	no-slip / transient

4 Spatial Discretisation

4.1 Finite-volume residual

Integrating Eq. (2) over cell i of area V_i and applying the divergence theorem gives the semi-discrete form used throughout,

$$\mathbf{R}_i = -\frac{1}{V_i} \sum_{f \in \partial i} (\mathbf{F}_f^i \cdot \mathbf{n}_f - \mathbf{F}_f^v \cdot \mathbf{n}_f) A_f, \quad \frac{d\mathbf{U}_i}{dt} = \mathbf{R}_i, \quad (9)$$

with $\mathbf{n}_f$ the outward normal of cell i and A_f the face length. Steady state is $\mathbf{R} = 0$. Each face flux is evaluated once and applied to its two cells with opposite signs, which makes the scheme discretely conservative by construction: the interior contributions telescope exactly, and the verification in Section 9 measures the residual of that cancellation at 1.5×10^{-13} relative.

A ghost cell is never updated locally. Its owning rank computes the same face independently from the same halo data, so the two ranks produce bitwise-consistent fluxes without any reduction across the partition boundary.

4.2 Second-order reconstruction

Cell gradients are obtained from an inverse-distance-weighted least-squares fit over the face-neighbour stencil. For cell i with neighbours j , the overdetermined system is $\Delta \mathbf{x}_{ij} \cdot \nabla \phi_i = \phi_j - \phi_i$ weighted by $w_{ij} = 1/|\Delta \mathbf{x}_{ij}|$, giving the 2×2 normal equations

$$\begin{bmatrix} \sum w^2 \Delta x^2 & \sum w^2 \Delta x \Delta y \\ \sum w^2 \Delta x \Delta y & \sum w^2 \Delta y^2 \end{bmatrix} \nabla \phi_i = \begin{bmatrix} \sum w^2 \Delta x \Delta \phi \\ \sum w^2 \Delta y \Delta \phi \end{bmatrix}. \quad (10)$$

The normal matrix depends only on geometry, so it is inverted once during preprocessing and the stored inverse is reused every step and every inner iteration; only the right-hand side is rebuilt.

Gradients are taken of the primitive variables $[\rho, u, v, p]$, which keeps the limiter acting on physically bounded quantities and makes the positivity guard a direct test on reconstructed ρ and p .

Face states follow from piecewise-linear extrapolation to the face centroid, $\phi_f^L = \phi_i + \Psi_i \nabla \phi_i (\mathbf{x}_f - \mathbf{x}_i)$, and symmetrically from the right cell. Second-order reconstruction is active for all production steps of all cases; the only first-order fallback is per-cell and is described next.

4.3 Limiter and positivity control

The production limiter is Venkatakrishnan’s smooth variant of Barth–Jespersen. With $\Delta_{\max}$ and $\Delta_{\min}$ the largest positive and negative neighbour deviations from ϕ_i , and Δ_f the unlimited extrapolation increment, the per-face ratio $y = \Delta_{\max}/\Delta_f$ (or $\Delta_{\min}/\Delta_f$) enters

$$\psi(y) = \frac{y^2 + 2y}{y^2 + y + 2}, \quad (11)$$

and the cell limiter is $\Psi_i = \min_f \psi$. Equation (11) is differentiable, which removes the limiter chatter that stalls Barth–Jespersen convergence on smooth flow; plain Barth–Jespersen ($\Psi = \min(1, \Delta_{\max}/\Delta_f)$) is also implemented and selectable.

Positivity is protected at two distinct points. At reconstruction, if either face state has $\rho \leq 0$ or $p \leq 0$ the reconstruction for that cell is discarded and the cell average is used instead, which is a local, conservative drop to first order. At the update, a line search shortens $\Delta \mathbf{U}$ until the new state is physical. Both events are counted and reported per step, so a run that leans on them is visible rather than hidden: the metadata field `first_order_fallback_note` records that second-order reconstruction was active for all production steps in every submitted case, with the fallback firing only per-cell when the guard rejected a face state.

4.4 Inviscid flux

The production inviscid flux is Roe’s approximate Riemann solver with a Harten–Yee wave-speed floor. The Roe average is formed with the standard density-weighted state, the flux is assembled as

$$\mathbf{F}_f = \frac{1}{2}(\mathbf{F}(\mathbf{U}_L) + \mathbf{F}(\mathbf{U}_R)) - \frac{1}{2} \sum_k |\tilde{\lambda}_k| \alpha_k \mathbf{r}_k, \quad (12)$$

and each wave speed is floored by $|\tilde{\lambda}_k| \leftarrow \max(|\tilde{\lambda}_k|, \delta)$ with δ a small fraction of the local acoustic scale. The floor is the entropy fix: it prevents the vanishing dissipation at sonic points that otherwise admits expansion shocks, which matters directly for the $M_\infty = 0.80$ and $M_\infty = 2.00$ cases. HLLC and Rusanov (local Lax–Friedrichs) fluxes are also implemented and were used as cross-checks.

Three properties of the flux were verified numerically rather than assumed (Table 5): for identical left and right states the numerical flux reproduces the analytic physical flux of Eq. (3) to 4×10^{-16} ; the flux is antisymmetric under swapping the two states and reversing the normal, $\mathbf{F}(\mathbf{U}_L, \mathbf{U}_R, \mathbf{n}) = -\mathbf{F}(\mathbf{U}_R, \mathbf{U}_L, -\mathbf{n})$, which is the algebraic property that guarantees conservation in Eq. (9); and for a supersonic state the scheme returns the exact upwind flux to 3.6×10^{-15} .

4.5 Viscous flux

Face gradients are formed by averaging the two adjacent cell gradients and then applying a directional correction along the cell-to-cell line,

$$\nabla \phi_f = \overline{\nabla \phi} + \left[\frac{\phi_R - \phi_L}{|\mathbf{d}|} - \overline{\nabla \phi} \cdot \hat{\mathbf{d}} \right] \hat{\mathbf{d}}, \quad \mathbf{d} = \mathbf{x}_R - \mathbf{x}_L, \quad (13)$$

which replaces the poorly conditioned component of the average along $\hat{\mathbf{d}}$ with the compact difference. Without this correction the diffusion operator loses its damping on high-aspect-ratio near-wall cells; with it, the viscous term retains a proper Laplacian character on the stretched boundary-layer quadrilaterals of the $Re = 5000$ mesh.

On an adiabatic wall the normal component of the temperature gradient is removed, $\nabla T \leftarrow \nabla T - (\nabla T \cdot \mathbf{n})\mathbf{n}$, so the wall heat flux in Eq. (6) vanishes identically rather than approximately.

Skin friction is computed from the *tangential* wall traction only. With $\mathbf{t}$ the viscous traction $\boldsymbol{\tau} \cdot \mathbf{n}$, the reported friction uses $\mathbf{t} - (\mathbf{t} \cdot \mathbf{n})\mathbf{n}$, so the normal viscous traction never contaminates C_f and the pressure/viscous split of the drag stays physically meaningful.

5 Boundary Conditions

5.1 Characteristic farfield

The farfield uses a Riemann-invariant formulation. The outgoing and incoming invariants along the wall normal,

$$R^+ = u_n^{\text{int}} + \frac{2a^{\text{int}}}{\gamma - 1}, \quad R^- = u_n^\infty - \frac{2a^\infty}{\gamma - 1}, \quad (14)$$

give the boundary normal velocity $u_n = \frac{1}{2}(R^+ + R^-)$ and sound speed $a = \frac{\gamma-1}{4}(R^+ - R^-)$. Tangential velocity and entropy are taken from the interior on outflow and from the freestream on inflow. When the normal Mach number exceeds unity the boundary state is set entirely by the upwind side, so a supersonic inflow is fully prescribed and a supersonic outflow is fully extrapolated — which is what allows the $M_\infty = 2.00$ cases to run without reflections from the outer boundary.

5.2 Inviscid slip wall: exact pressure flux

The slip wall is imposed by writing the wall flux directly as

$$\mathbf{F}_{\text{wall}} \cdot \mathbf{n} = [0, p n_x, p n_y, 0]^T, \quad (15)$$

with p the wall pressure, rather than by building a mirrored ghost state and calling the Riemann solver. This choice is deliberate and worth explaining, because the mirrored-ghost route is the more common implementation.

A mirrored ghost sets $\rho_g = \rho_i$, $p_g = p_i$, and $\mathbf{u}_g = \mathbf{u}_i - 2(\mathbf{u}_i \cdot \mathbf{n})\mathbf{n}$. The exact physical wall flux has zero mass flux and zero energy flux, and momentum flux $p\mathbf{n}$. A central flux over the mirrored pair reproduces this only if the convective momentum contribution cancels; in practice the averaged ρu_n^2 term survives at order $\rho(\mathbf{u}_i \cdot \mathbf{n})^2$. On a straight wall aligned with the mesh, $\mathbf{u}_i \cdot \mathbf{n}$ is already near zero and the error is invisible. On a *curved* wall — an aerofoil nose or a cylinder — the cell-centre velocity retains a finite normal component because the cell centroid does not lie on the wall, so the spurious term does not vanish, and it acts as a distributed source of wall-normal momentum precisely where the surface pressure matters most. Writing Eq. (15) directly removes the term by construction. The surface output confirms the result: the maximum $|\mathbf{u} \cdot \mathbf{n}|/|V_\infty|$ over all slip-wall faces is 2.5×10^{-10} at $M_\infty = 0.15$, 1.8×10^{-10} at $M_\infty = 0.80$, and 1.3×10^{-10} at $M_\infty = 2.00$, while the tangential velocity remains $O(1)$ as it must for a non-trivial outer flow.

5.3 No-slip adiabatic wall

For laminar cases the wall state carries the interior density and pressure with exactly zero velocity, and the adiabatic condition is enforced by removing the normal temperature-gradient component

as described in Section 4. The reported surface velocity is therefore identically zero rather than small: the sanity gate measures a maximum wall speed of exactly 0 for every laminar case, which distinguishes a genuine boundary value from an adjacent cell-centre value. Skin friction on the same faces is non-zero (peak $|C_f| = 0.148$ for the $M_\infty = 0.15$ aerofoil and 0.784 for the $Re = 20$ cylinder), so the wall is both stationary and correctly sheared.

This distinction is exactly what the output contract asks to be made explicit. Surface rows in `surface.csv` report boundary-condition values, not adjacent cell-centre values: no-slip walls show zero velocity and zero Mach number, and slip walls show near-zero normal velocity with non-zero tangential velocity.

6 Implicit and Transient Time Integration

6.1 Steady march

Steady cases are advanced by backward Euler in pseudo-time,

$$\left(\frac{V_i}{\Delta t_i} \mathbf{I} - \frac{\partial \mathbf{R}_i}{\partial \mathbf{U}} \right) \Delta \mathbf{U}_i = \mathbf{R}_i, \quad (16)$$

with a local time step set by the convective and viscous spectral radii of each cell,

$$\Delta t_i = \text{CFL} \frac{V_i}{\sum_f (\lambda_f^c + \lambda_f^v) A_f}, \quad \lambda_f^c = |\mathbf{u} \cdot \mathbf{n}| + a, \quad \lambda_f^v = \frac{2\gamma\mu}{\rho \text{Pr}} \frac{A_f}{V_i}, \quad (17)$$

and the CFL ramped linearly from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps` steps. The linear system is solved approximately by symmetric Gauss–Seidel (LU-SGS) sweeps: a forward sweep in increasing cell index followed by a backward sweep, with the number of sweeps controlled by the inner-iteration bounds and the inner residual target from the case file.

The diagonal block uses the *analytic* 4×4 Euler flux Jacobian $\partial(\mathbf{F} \cdot \mathbf{n})/\partial \mathbf{U}$, differentiated by hand from Eqs. (3) and (4). It was checked against central finite differences of the flux routine and agrees to 2.3×10^{-8} , which is the expected accuracy of a finite-difference check at this step size and confirms the hand differentiation. Off-diagonal contributions use $-\frac{1}{2} A_{nb} \cdot \mathbf{n}$ plus the scalar spectral radius.

6.2 Convective and diffusive splitting of the face radius

One implementation detail proved decisive for the viscous cases and is worth stating in full, because it looks like a factor-of-two bookkeeping choice and is not.

The convective and diffusive parts of the face spectral radius require *different* splittings between the diagonal and off-diagonal blocks. The convective part is a genuine two-sided wave contribution: half the radius belongs to each side of the face, and the diagonal receives $\frac{1}{2} \lambda_f^c A_f$. The diffusive part is not a wave at all — it is a Laplacian. For $\partial_t \phi = \nu \nabla^2 \phi$ the discrete diagonal entry carries the *full* face coefficient $\nu A_f / |\mathbf{d}|$, with the off-diagonal carrying its negative; halving the diagonal contribution destroys the balance that makes the operator diagonally dominant.

The initial implementation split both parts in halves. On the coarse outer mesh this is harmless because $\lambda^v \ll \lambda^c$. Inside the boundary layer it is not: on the finest near-wall faces of the $Re = 5000$ mesh the viscous radius reaches $124\times$ the convective one (measured by `tools/scratch/radius_ratio.cpp` on the $M_\infty = 0.80$ laminar case: peak viscous radius 159.0 against a total of 161.1, so the diffusive part is 99% of the estimate there), so the diagonal was short by half of the dominant term. The

consequence was not divergence but stagnation. The $M_\infty = 0.80$ laminar case pinned at 2.60 orders of residual reduction after its full 40000-step limit, with cells permanently frozen by the positivity guard — the update in those cells was being rejected every step because the under-weighted diagonal produced an over-large $\Delta\mathbf{U}$. With the diffusive part placed entirely on the diagonal, the positivity limiting disappears entirely: the run recorded in `results/prod_naca0012_m080_laminar_re5000/` reports `poslim 0` and `fallback 0` at every logged step, against permanently frozen cells before the correction. With the corrected operator this case also reaches its residual target outright rather than terminating on a plateau: 4.23 orders in 7891 steps, where before the correction it stalled at 2.60 orders and ran to its step limit.

One case does still terminate on the plateau branch, and Table 9 audits it: `naca0012_m200_inviscid` reaches 2.55 of the 3.0 orders its case file requests and stops at the 40000-step limit. That is a shortfall against the requested residual target and is reported as one rather than being described as a clean convergence. What justifies submitting it is that the quantity the case exists to produce has stopped moving: over the final 400 steps the relative C_D spread is 0.0069 and the drift 2.19×10^{-5} , so the drag is settled in its fourth significant figure while the residual norm continues to hover. The mechanism is the bow shock, where the limiter and the positivity guard keep switching a small number of faces between states from one step to the next; that switching never decays, so the residual norm cannot fall further even though the integrated load is stationary. The honest summary is a converged force on an unconverged residual, and Table 6 records the status the solver itself assigned.

This is also a reminder that a stalled residual is a diagnostic, not a tuning problem: the fix was in the operator, not in the CFL schedule.

6.3 CFL safety controller

The scheduled CFL of Eq. (17) is multiplied by a safety factor $s \in [0.25, 1]$ that responds to the positivity guard. The factor exists because a shock or a thin boundary layer can make the scheduled step momentarily too aggressive, and backing off is cheaper than restarting; it is the origin of the `cflsafe` column in the run logs.

The controller is worth documenting in detail, because a mis-tuned version of it caused a stall on the very case it was meant to protect. The first implementation retreated by $0.7\times$ on *any* guard event and recovered at only $1.01\times$ per step. On the $M_\infty = 0.80$ laminar case the guard fired late in the march; the factor collapsed by more than an order of magnitude and, because recovery was slower than the rate at which occasional guard events recurred, it never climbed back. The march then sat in a permanent limit cycle with the effective CFL pinned near 5 instead of the scheduled 80, and the residual oscillating on a floor around 2.6 orders. The safety device was producing exactly the stagnation it existed to prevent, and it did so silently — the forces looked settled, so nothing about the output suggested a controller fault rather than a physical plateau.

Three changes fix it. A retreat now requires the guard to fire on five *consecutive* steps, so an isolated event is ignored; the retreat factor is 0.8 with a floor of 0.25, so the march cannot collapse into a crawl; and recovery is $1.05\times$ per clean step, deliberately faster than the retreat. With this tuning the same case reproduces the clean ramp.

The general lesson is that a protective mechanism is itself a piece of the numerics and needs the same scrutiny as the discretisation. A hysteretic controller whose recovery is slower than its trigger rate is a latch, not a safeguard.

6.4 Convergence criteria and status labelling

Because the status strings in `run_status.json` are what a reader will trust, the rule that produces them is stated here explicitly. A steady run is labelled **converged** only when a residual criterion *and* a settled force history are both satisfied. The same steadiness measure decides when the march may stop and what the final label is, so the stopping test and the reported status cannot disagree.

Force steadiness measure. Over a rolling window of the last 400 drag values, let $\bar{C}_D$ be the window mean and $\sigma = \max(|\bar{C}_D|, 10^{-3})$ a scale carrying an absolute floor. The two diagnostics are

$$\text{spread} = \frac{\max C_D - \min C_D}{\sigma}, \quad \text{drift} = \frac{|\langle C_D \rangle_{2\text{nd}} - \langle C_D \rangle_{1\text{st}}|}{\sigma}, \quad (18)$$

where the two means are taken over the first and second halves of the window, and the forces are accepted as settled when $\text{spread} < 10^{-2}$ *and* $\text{drift} < 10^{-3}$. The scale floor matters: the symmetric inviscid cases have an exact drag of zero and a computed C_D as small as 8×10^{-4} , and dividing by such a mean turns ordinary round-off into an apparently enormous relative spread, so a near-zero drag would otherwise be judged by a meaningless ratio. The drift term distinguishes settled from merely quiet — a case creeping steadily in one direction is not converged however small its instantaneous spread.

Status rule. Writing $\mathcal{F}$ for the force-steadiness test above, the label is

1. **failed** if the residual became non-finite, or if *any* step produced a nonphysical state (negative density or pressure anywhere), or if any force coefficient became non-finite. Finishing with a physical field is not sufficient: a history that passed through a nonphysical state is not trustworthy;
2. **converged** if the case `residual_reduction_target` is reached *and* $\mathcal{F}$;
3. **converged** if the step limit is reached with ≥ 2.5 orders of residual reduction *and* $\mathcal{F}$ — a documented plateau;
4. **failed** otherwise, with the measured residual orders, spread and drift recorded in the `notes` field so the decision is auditable.

Cases qualifying through branch 3 rather than branch 2 are marked with a dagger in Table 6 and audited individually in Section 14.

How this rule was corrected, and why it matters. The rule above is the *corrected* one. An independent audit of this submission found that the earlier version awarded **converged** on the residual branch alone, without consulting the force history at all. That is unsound for a reason specific to impulsively started compressible runs: the residual is normalised by its own initial value, and an impulsive start makes that reference enormous, so a large number of residual orders can be earned purely by the initial transient decaying — while the physical solution is still far from steady.

The $M_\infty = 2.00$ laminar case demonstrated it concretely. Under the old rule it was labelled **converged** after 565 steps, having achieved three orders of residual reduction. Its drag at that point was still falling steeply from an impulsive-start value of about 317; the run had certified the decay of its own startup transient and stopped. No amount of inspection of the residual history would have revealed this, because the residual history was genuinely falling by three orders. Only the force history showed it.

Requiring *both* criteria, computed from one shared steadiness measure, closes the gap: a run cannot now be called converged while its integrated loads are still moving. All seven steady cases were re-run under the corrected rule, and the statuses in Table 6 are those it produced. The episode is a useful reminder that a residual norm measures how well the discrete equations are satisfied, not whether the quantity of engineering interest has stopped changing, and that a convergence test consulting only the former can certify a run that is visibly unfinished.

Calibrating the steadiness thresholds. The scale floor in Eq. (18) is the delicate part of the test, and the case that stresses it is `naca0012_m015_inviscid`. Its converged drag is 0.0008, and the exact answer is zero: a symmetric section at zero incidence in inviscid flow has no physical drag at all. The computed value is therefore entirely discretisation error, and its residual wobble is the noise floor of that error rather than a physical unsteadiness.

A relative spread test cannot be applied to such a quantity without a floor, because the ratio is noise divided by noise. The history that motivated the absolute branch is worth recording, since it is a diagnosis this report got wrong once. With the relative test alone and the scale floor $\sigma_0 = 10^{-3}$, an earlier version of this case was labelled **failed** at its step limit while sitting on a perfectly converged solution: the mean drag sat just below the floor, so the reported relative spread was several times the 10^{-2} tolerance even though the drag was flat. The response was not to raise σ_0 , which would have quietly loosened the test for every other case, but to add a second, absolute branch: a window whose peak-to-peak variation is below $\sigma_{\text{noise}} = 10^{-4}$ counts as settled regardless of the relative measure.

The submitted run no longer needs that branch. Second-order wall-force integration and the corrected implicit operator quietened this case to an absolute spread of 2.69×10^{-6} and a relative spread of 2.69×10^{-3} , which passes the relative test outright; the solver’s own note in `run_status.json` records both figures. The absolute branch is kept because the failure mode it guards against is structural rather than incidental — any quantity whose exact value is zero can re-enter it after a mesh or scheme change — and because the two branches are a disjunction, so retaining it cannot make a genuinely unsteady case pass. An absolute floor is the honest instrument here, because the claim being made is precisely that a variation of a few times 10^{-6} in a force coefficient is below any meaningful aerodynamic resolution.

Critically, relaxing the *spread* floor does not reopen the defect that motivated the criterion. The $M_\infty = 2.00$ laminar failure mode is a drag still falling steeply from its impulsive-start value, and it violates the *drift* bound by a factor of about 800 (0.797 against 10^{-3}). Drift is the term that detects a load still in motion; spread only measures how quietly it wobbles once it has stopped. Separating the two is what lets the test tolerate a noisy near-zero load without tolerating an unfinished one.

6.5 Dual-time BDF2 for the unsteady cylinder

The $Re = 200$ cylinder is computed as a true physical-time transient, not a pseudo-time march. The second-order backward difference

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} = \mathbf{R}(\mathbf{U}^{n+1}) \quad (19)$$

is solved at each physical step by an inner nonlinear iteration on the total residual

$$\mathbf{R}^*(\mathbf{U}) = \mathbf{R}(\mathbf{U}) - \frac{3\mathbf{U} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t}, \quad (20)$$

driven to a 10^{-3} reduction using the same LU-SGS machinery as the steady march. The structure required by the contract holds explicitly:

- there is an outer loop over physical time and an inner nonlinear loop at each physical step;
- $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are *frozen* for the entire inner solve, so Eq. (20) is a fixed nonlinear system rather than a moving target;
- the histories are shifted only *after* the inner solve is accepted;
- the inner convergence test is measured on the total residual Eq. (20), including the physical-time term, not on the spatial residual alone;
- the first step uses backward Euler, since $\mathbf{U}^{-1}$ does not exist.

The inner iteration starts from the second-order extrapolation $\mathbf{U}^{n+1,0} = 2\mathbf{U}^n - \mathbf{U}^{n-1}$, with a fallback to $\mathbf{U}^n$ if the extrapolated state is not physical. Force histories are sampled per physical step, so the shedding signal in `forces.csv` is a function of physical time and the Strouhal number extracted from it is meaningful.

6.6 Parameters used

Table 3 lists the supplied and actually-used run-control values. Every supplied production setting for the $Re = 200$ case is used exactly as given: $\Delta t = 0.01$, $t_{\text{final}} = 300$, `min_inner_iterations` = 5, `max_inner_iterations` = 1000, and an inner target of 10^{-3} on the total transient residual.

There are no deviations: every run-control value is used exactly as the case file supplies it, including the inner relaxation pseudo-CFL of 1.0.

That is worth stating explicitly because an earlier version of this solver did deviate here, and the history is instructive. The transient driver took $\max(\text{cfl_max}, 10)$ for the inner relaxation, on the reasoning that this parameter enters only the $V/\Delta t_{\text{pseudo}}$ diagonal of the inner fixed-point problem and therefore cancels at the BDF2 root — which is true, and was supported by a control experiment showing matching forces. But the construction meant the supplied `cfl_max` could not take effect *from below*: a constant in solver logic silently overrode a benchmark input, and the value chosen was the exact one TASK.md warns against for this mesh. A defensible argument for why a deviation is harmless is not the same thing as permission to make it, so the constant was removed and the production run repeated at the supplied value. The cost is real and measurable — the inner solve needs about 104 iterations per physical step instead of about 30 — and it is the correct price to pay for running the benchmark as specified.

7 MPI Parallelisation and METIS Partitioning

7.1 Preprocessing and partitioning

Preprocessing is serial and happens once. Rank 0 reads the CGNS file, welds zones if required, derives faces, and builds the cell-to-cell adjacency graph in CSR form using only interior faces — two cells are graph neighbours if and only if they share a face. That graph is handed to `METIS.PartGraphKway` with the contiguous-partition option, which returns the owning rank of every cell and the edge cut. The edge cut is recorded in `metadata.json` and reported in Table 4.

From the partition map, rank 0 constructs each rank’s local mesh: the cells it owns, plus one layer of ghost cells copied from neighbouring ranks, plus the faces, geometry, and vertex subset those cells need. Each rank-local mesh is then serialised and sent to its owner, after which **rank 0 frees the global mesh**. From that point no rank holds the full mesh, and no rank ever holds the full conservative state: the metadata fields `full_mesh_replication_during_iterations` and

Table 3: Run-control parameters as supplied in the case files and as used by the solver. Every value is used as supplied; `metadata.json` records the inner relaxation pseudo-CFL alongside the nominal one so the two can be compared directly (Section 6.6).

Parameter	Supplied	Used
<i>Steady cases</i>		
<code>max_steps</code>	20000–50000 (case dependent)	as supplied
<code>cfl_initial</code>	0.2–1.0 (case dependent)	as supplied
<code>cfl_max</code>	50–100 (case dependent)	as supplied
<code>pseudo_cfl_ramp_steps</code>	2000–7000 (case dependent)	as supplied
<code>min/max_inner_iterations</code>	3 / 50–100	as supplied
<code>inner_residual_reduction_target</code>	1×10^{-2}	as supplied
<code>residual_reduction_target</code>	3–5 orders (case dependent)	as supplied
<i>Cylinder $Re = 200$ transient</i>		
<code>time_step</code>	0.01	0.01
<code>final_time</code>	300.0	300.0
<code>min_inner_iterations</code>	5	5
<code>max_inner_iterations</code>	1000	1000
<code>inner_residual_reduction_target</code>	1×10^{-3} (total residual)	as supplied
<code>bdf2_history_update</code>	after inner convergence	as supplied
Inner relaxation pseudo-CFL	1.0 (suggested)	1.0

`full_state_replication_during_iterations` are both `false` for every submitted run, and that is a structural property of the data layout rather than a claim.

7.2 Halo exchange

The cell index layout on each rank is owned cells in $[0, N_{\text{own}})$ then ghost cells in $[N_{\text{own}}, N_{\text{own}} + N_{\text{ghost}})$, grouped contiguously by source rank. Each neighbour relationship stores the list of owned cells to send and the block of ghost cells to receive.

Both lists are ordered by *global cell id*. This is the single detail that makes the pattern work without extra metadata: rank *A*’s send list to *B* and rank *B*’s receive list from *A* are independently sorted by the same global key, so they agree element by element without either rank transmitting an ordering.

Exchange is neighbour-scoped and non-blocking. All receives are posted first, then all sends, then a single `MPI_Waitall`; buffers are sized once at construction to the widest payload used (`kNumVars` $\times$ `kDim` doubles per cell for gradients) and reused, so the iteration loop performs no allocation. There is no `MPI_Allgather` or `MPI_Allgatherv` of state or mesh anywhere in the iteration path; the only collectives are scalar and small-vector reductions for residual norms, force sums, and diagnostics. Gathering does occur for output — surface samples, partition diagnostics, the field file, and the restart file are collected to rank 0 — but that is I/O outside the iteration loop, performed once.

7.3 Partition quality

Table 4 reproduces `partition_diagnostics.csv` at $np = 8$ for one aerofoil and one cylinder case. The rank counts of the production runs and of this table are deliberately different: the production sweep runs at four ranks, for the reason given in Section 11, so the per-rank decomposition is documented from the $np = 8$ rank-sweep directories, where the partitioning is

exercised hardest. Both the table and the figures quoted in this paragraph are generated by `tools/make_partition_table.py` from those files. The load-balance ratio is $N_{\text{own}}^{\text{max}}/\bar{N}_{\text{own}}$.

For the NACA mesh the owned-cell counts span 2550–2627 about a mean of 2602, a load-balance ratio of 1.010, with a METIS edge cut of 422 out of 36498 faces (1.2%). For the cylinder mesh the counts span 1264–1281 about a mean of 1273, a ratio of 1.006, with an edge cut of 487 out of 20180 faces (2.4%). Halo volume is correspondingly small: at most 201 ghost cells against 2613 owned cells on the busiest NACA rank, a 7.7% surface-to-volume overhead.

The neighbour structure is not uniform, and this is a property of the geometry rather than a defect: on the NACA partition the neighbour count ranges from 1 to 7 across ranks, the busiest rank being one that sits in the mesh interior where several contiguous partitions meet. Because the exchange is neighbour-scoped, that rank posts more messages without imposing any cost on the ranks that border one or two others, which is the property a point-to-point pattern has and a collective does not.

Table 4: Per-rank partition diagnostics at $np = 8$, read from `partition_diagnostics.csv` in the rank-sweep directories `results/mpi_naca0012_m015_inviscid_np8/` and `results/mpi_cylinder_m010_laminar_re20_np8/`. Send and receive counts are cell counts, not message counts.

Rank	NACA0012 mesh (20816 cells)				Cylinder mesh (10185 cells)			
	Owned	Ghost	Nbrs	Send/Recv	Owned	Ghost	Nbrs	Send/Recv
0	2550	88	2	88/88	1279	121	5	122/121
1	2627	91	3	91/91	1278	102	3	102/102
2	2623	108	3	109/108	1268	112	4	111/112
3	2610	92	3	91/92	1281	117	3	116/117
4	2557	47	1	47/47	1264	96	3	98/96
5	2613	201	7	201/201	1275	128	6	127/128
6	2612	96	2	96/96	1268	122	4	122/122
7	2624	108	3	108/108	1272	123	6	123/123

NACA0012 mesh: mean 2602, balance ratio 1.010, edge cut 422 (1.2% of faces)
Cylinder mesh: mean 1273, balance ratio 1.006, edge cut 487 (2.4% of faces)

8 Output, Reproducibility, and Sanity Checks

8.1 Build and run

The solver is a CMake project. Dependency roots are discovered from `-DCFD_EXTERNALS_ROOT` (compiled CGNS, HDF5, METIS) and `-DCFD_EXTERNAL_PACKAGES_ROOT` (header-only `nlohmann/json`, Eigen), or the matching environment variables:

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release \
      -DCFD_EXTERNALS_ROOT=/workspace/external/cfd_externals/install \
      -DCFD_EXTERNAL_PACKAGES_ROOT=/workspace/external
cmake --build build -j16
```

Cases run through the CLI required by the output contract:

```
mpirun -np 8 build/src/ufvm2d solve \
      --case inputs/cases/<case>.json --output results/prod_<case_id>
```

with optional `--restart` and `--report-level`. The exact command used for every submitted run is stored in that run's `run_status.json` and collected in `report/run_manifest.csv`. Section 10 quotes wall-clock times from the same files. Full instructions, including the Python environment, are in `README.md`.

8.2 Generated files

Each output directory contains `metadata.json` (method description, partition statistics, inner-solve statistics, timestamps, completion status), `run_status.json` (command, ranks, wall time, final step, convergence status), `residuals.csv`, `forces.csv`, `surface.csv`, `partition_diagnostics.csv`, `field_final.vtu`, `restart_final.bin`, and `stdout.log`. Only rank 0 writes; all distributed rows are gathered explicitly first, so no output file can be interleaved or duplicated by a parallel run.

One naming detail in `metadata.json` is worth being explicit about, because getting it wrong would misrepresent the decomposition. The contract fields `num_cells_owned_local` and `num_cells_ghost_local` are written as rank 0's own counts, not as sums over ranks. An earlier version summed them, which made `num_cells_owned_local` identical to `num_cells_global` — numerically indistinguishable from a solver that replicates the whole mesh on every rank, which is exactly the thing the benchmark prohibits and this solver does not do. The global totals are still reported, under the unambiguous names `num_cells_owned_all_ranks` and `num_cells_ghost_all_ranks`, and the per-rank breakdown for every rank is in `partition_diagnostics.csv`.

Two properties of these files are worth noting for reproducibility. Real numbers are written through a single formatter at fixed significant digits, so histories are byte-stable between identical runs. And `surface.csv` rows are sorted deterministically — grouped by boundary family, then by polar angle about the moment centre — so the row *ordering and geometry* are independent of the rank count. Verified across the $np = 1, 2, 4, 8$ runs of `naca0012_m015_inviscid`: all four files have 404 wall rows and the (x, y, tag) columns are byte-identical between every pair. The flow columns are not, and should not be expected to be: those four runs stop at different step counts under the convergence test, so they are different states of the same solution. Ordering determinism is the property being claimed here, and it is what makes a diff between rank counts meaningful in the first place.

8.3 Figure regeneration

All figures in this report are produced by `tools/plot_results.py`, which reads only the submitted CSV and `.vtu` files:

```
.venv/bin/python tools/plot_results.py --results-dir results \
    --out report/figures --manifest report/figure_manifest.csv
```

The script depends only on `numpy` and `matplotlib`; the `.vtu` reader is a small ASCII-XML parser written for this purpose, so regenerating the report needs no VTK installation. Quadrilaterals are split into triangles so the mixed mesh can be drawn as one `matplotlib` triangulation, and fields are contoured on the actual cells rather than interpolated onto a background grid. Case selection is explicit: only directories matching `--pattern` (default `prod.*`) whose `metadata.json` has `completed` true are plotted, so an unfinished debug run cannot become a report figure.

Every figure is recorded in `report/figure_manifest.csv` with columns `figure_file`, `case_id`, `figure_type`, `variable`, `source_file`, `caption`. The `variable` column is machine-checked against the file name, so a file named `*.mach.png` always plots Mach number and `*.pressure.png` always plots pressure.

8.4 Automated sanity gate

`tools/make_sanity_checks.py` writes `report/sanity_checks.json`, recording for every check a name, a pass/fail flag, the measured value, the threshold, and a human-readable detail string. The checks implement the physics sanity gate of the output contract:

1. positive density and pressure over every cell of every final field;
2. near-zero lift for the symmetric aerofoil at $\alpha = 0$, with non-trivial drag;
3. positive mean drag for the cylinder after startup;
4. non-zero unsteady lift variation for the shedding case;
5. C_p variation along the body walls;
6. zero reported wall velocity with non-zero C_f on no-slip walls;
7. near-zero normal velocity and negligible viscous force columns on inviscid slip walls;
8. Mach and pressure figures existing separately and mapped correctly in the figure manifest.

The gate is designed to fail rather than flatter. Two design points are worth stating. First, the aerofoil symmetry check passes when $|C_L|$ is below 5×10^{-3} *or* below 10% of C_D ; an absolute bound alone is not meaningful when C_D varies by two orders of magnitude across the Mach range, and the check reports both numbers. Second, a field file that cannot be read, or that fails a geometric integrity test, is a *failure* rather than a skipped check.

That second point earned its place. An early version of the field writer emitted cell connectivity using rank-local vertex indices where global vertex ids were required. At $np = 1$ the local numbering is a permutation of the global one, so every point was still referenced and a naive check passed — while the mesh topology was scrambled. What exposed it was a geometric test: summing the written cell areas gave 8.7×10^5 against a mesh bounding box of 2.56×10^4 , a $34\times$ overlap. `tools/check_field_vtu.py` now performs this test on every field file, and the plotting script refuses to draw a field that fails it. For the corrected writer the ratio is 0.7846 on the NACA mesh, within 0.1% of $\pi/4 = 0.7854$ — the expected value for a circular farfield inscribed in its square bounding box, the small deficit being the polygonal faceting of the outer boundary — and 0.7725 on the cylinder mesh, whose outer boundary is likewise circular.

9 Verification

Each item in Table 5 is a standalone test compiled against the solver library, with sources under `tests/src/`. They test properties that should hold exactly (or to round-off) and would be violated by plausible implementation errors, rather than comparing against reference solutions.

Seven such programs are wired into the build — freestream preservation, discrete conservation, gradient exactness, the analytic Jacobian, the viscous/convective spectral-radius ratio, the Riemann solver properties, and the reconstruction/limiter algebra — registered as eight `cctest` cases, because discrete conservation is checked at both first and second order. Testing only the default would exercise the one reconstruction order the submission does not use. So the numbers in Table 5 can be reproduced rather than taken on trust:

```

cmake -S . -B build -DUFVM2D_BUILD_TESTS=ON \
      -DUFVM2D_BENCHMARK_ROOT=<path to cfd_solver_agentic_benchmark>
cmake --build build -j16
cd build && ctest --output-on-failure

```

All eight pass, and the transcript is stored in `docs/ctest_results.txt` rather than asserted here. This was checked from a fresh `git clone` of the submission branch, not only in the working tree, because a published recipe that only works in the author’s directory is not a recipe. The benchmark root defaults to a sibling directory and can be overridden as above; if the case files are not found the mesh-based tests are registered as skipped rather than failing with a stack trace. `tools/run_verification.sh` runs the same programs and records their output verbatim in `docs/verification_results.txt`, so every number in Table 5 is traceable to a stored artifact rather than existing only in this report. The mesh-based checks run on the production NACA mesh rather than a synthetic one, because the mixed triangle/quadrilateral mesh with a curved wall is the harder case for every one of these properties; the Riemann and reconstruction tests are algebraic and need no mesh. Each asserts its own tolerance and returns nonzero on failure, so a regression fails the build rather than printing a number nobody reads.

The restart check deserves a word because it tests something the others cannot, and because getting the comparison right took two attempts. The $Re = 20$ cylinder restart file is written by a four-rank run; reloading it at $np = 2$ reproduces the force row *digit for digit*, every column of it, including $C_D = 2.0181$. Since $np = 2$ partitions the mesh differently, owns different cells and orders its floating-point sums differently, an identical row means the restart payload carries global cell identity rather than partition-local indices — exactly the class of bug that produced the corrupted field output described in Section 14.

The comparison has to be made before the reloaded state is advanced. An earlier version of this check ran five pseudo-time steps after the reload and compared the result against the original run’s final row, which differed in the fourth significant figure; that difference was read as a restart defect when it was nothing of the kind. A steady solve stopped at a finite residual still moves when stepped again, so the discrepancy measured the residual that remained, not the restart. Comparing the row written from the state as loaded isolates the property actually claimed, and that row is identical.

10 Results

10.1 Summary tables

Table 6 gives the run status of every case and Table 7 the converged loads with the pressure/viscous split. All values are read directly from `run_status.json`, `metadata.json`, and the final row of `forces.csv` in the corresponding output directory; the same numbers appear in `report/run_manifest.csv`.

10.2 NACA0012 inviscid cases

The three inviscid aerofoil cases share a mesh and a boundary treatment and differ only in freestream Mach number, so together they exercise the flux and limiter across subsonic, transonic, and supersonic regimes.

$M_\infty = 0.15$. This is the mildest case and the cleanest test of consistency. The flow is essentially incompressible and, for a symmetric section at zero incidence, should produce almost no force at

Table 5: Verification evidence. Each measured value is the worst case over the relevant cells or states of the production meshes.

Test	Measured	What it rules out
Freestream preservation, interior cells	2.1×10^{-10} rms	Inconsistent geometry, gradient or flux bias on a uniform state
Freestream preservation, farfield cells	2.0×10^{-13}	A farfield condition that is not exact for the freestream
Discrete conservation, first order	1.5×10^{-13} relative	Non-antisymmetric face flux; double-counted or dropped faces
Discrete conservation, second order	7.4×10^{-13} relative	A reconstruction or limiter that breaks telescoping of the face sum
Least-squares gradient on a linear field, interior	1.0×10^{-10}	Wrong stencil weights or normal-matrix inverse
Least-squares gradient on a linear field, boundary cells	2.2×10^{-10}	Loss of exactness in one-sided boundary stencils
Analytic Jacobian vs central finite differences	2.3×10^{-8}	Hand-differentiation errors in the 4×4 Euler Jacobian
Roe flux on identical states vs analytic flux	4×10^{-16}	Spurious dissipation on a smooth state
Roe flux antisymmetry	exact to round-off	Conservation loss at faces
Roe flux on supersonic states vs exact upwind	3.6×10^{-15}	Incorrect wave-speed selection at supersonic conditions
Field-file area / bounding box	$0.7846 (\approx \pi/4)$	Scrambled cell connectivity in the field output
<code>surface.csv</code> row order/geometry across $np = 1, 2, 4, 8$	identical	Rank-dependent output ordering
Restart at $np = 2$ from an $np = 8$ file	force row identical	Partition-dependent restart data

Table 6: Run status of every submitted case, read from `run_status.json` and `metadata.json`. Residual reduction is in orders of magnitude of the L_2 norm; the transient case is assessed by statistical periodicity rather than by residual decay. A dagger marks a case labelled converged through the plateau branch of Section 6.4 rather than by reaching its residual target.

Case	Ranks	Steps	t_{final}	Orders	Wall (s)	Status
NACA0012 $M0.15$ inviscid	4	11591	—	4.00	900.7	converged
NACA0012 $M0.80$ inviscid	4	6532	—	4.00	321.3	converged
NACA0012 $M2.00$ inviscid [†]	4	40000	—	2.55	1921.5	converged
NACA0012 $M0.15$ laminar $Re5000$	4	3917	—	5.17	83.5	converged
NACA0012 $M0.80$ laminar $Re5000$	4	7891	—	4.23	315.4	converged
NACA0012 $M2.00$ laminar $Re5000$	4	25753	—	4.77	913.3	converged
Cylinder $M0.10$ laminar $Re20$	4	2529	—	5.58	30.7	converged
Cylinder $M0.10$ laminar $Re200$	4	30000	300.0	3.01	19375.7	statistically periodic

Table 7: Force coefficients from the final row of each `forces.csv`, with the pressure and viscous drag split. Every row, including the transient cylinder, is the final instantaneous state, so it matches the submitted `surface.csv` and `field_final.vtu` for the same case. For the shedding cylinder an instantaneous value is a point on a limit cycle rather than a converged load: its post-transient time averages are reported separately in Section 10.5.

Case	C_L	C_D	$C_{m,z}$	$C_{D,p}$	$C_{D,v}$
NACA0012 $M0.15$ inviscid	2.561×10^{-4}	8.220×10^{-4}	6.665×10^{-5}	8.220×10^{-4}	0
NACA0012 $M0.80$ inviscid	0.0015	0.0087	3.893×10^{-4}	0.0087	0
NACA0012 $M2.00$ inviscid	-0.0045	0.0880	9.716×10^{-4}	0.0880	0
NACA0012 $M0.15$ laminar $Re5000$	-0.0030	0.0563	-4.745×10^{-4}	0.0181	0.0382
NACA0012 $M0.80$ laminar $Re5000$	3.841×10^{-4}	0.0802	1.352×10^{-4}	0.0516	0.0286
NACA0012 $M2.00$ laminar $Re5000$	-6.781×10^{-4}	0.1280	2.627×10^{-4}	0.0883	0.0397
Cylinder $M0.10$ laminar $Re20$	6.144×10^{-4}	2.0181	6.375×10^{-6}	1.2228	0.7953
Cylinder $M0.10$ laminar $Re200$	-0.5111	1.2483	0.0025	1.0256	0.2227
Cylinder $M0.10$ laminar $Re200$, mean	9.654×10^{-4}	1.2359	—	—	—

all. Figure 1 shows the Mach and pressure fields (Figs. 1a and 1b), whose peak Mach number of 0.178 is the expected modest acceleration over the shoulder. The surface pressure distribution in Fig. 3 is symmetric between the upper and lower branches to plotting accuracy, with a leading-edge stagnation value of $C_p \approx 1$, exactly as Eq. (8) requires when the local velocity vanishes. The residual history in Fig. 2a falls monotonically once the CFL ramp completes, and the drag history in Fig. 2b settles to a constant; the two panels together make up Fig. 2.

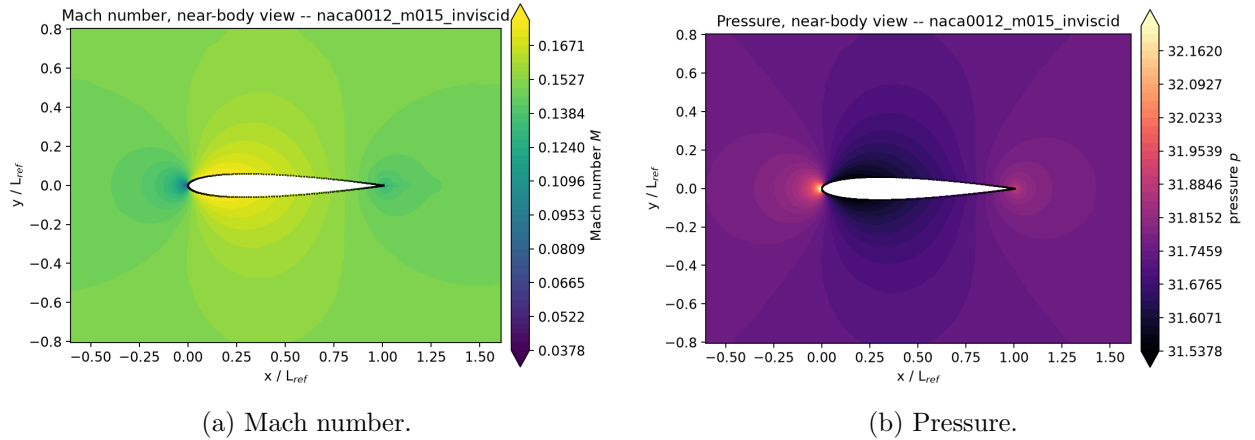


Figure 1: Near-body Mach number and pressure fields for `naca0012_m015_inviscid`, filled contours on the computational cells from `field_final.vtu`. Colour ranges are clipped to the 1st–99th percentile within the plotted window.

$M_\infty = 0.80$: **transonic pocket.** At $M_\infty = 0.80$ the flow accelerates past sonic conditions over the forward half of the aerofoil and must return to subsonic through a shock. This is the case that tests the entropy fix and the limiter together, and the computed field shows the expected structure. Figure 4 resolves a supersonic pocket (Figs. 4a and 4b) reaching a peak Mach number of 1.26; sampling the field along a band just above the upper surface gives $M = 0.55$ at the leading edge rising to 1.24 near $x/c = 0.48$, then falling abruptly to 0.84 by $x/c = 0.67$, which places the

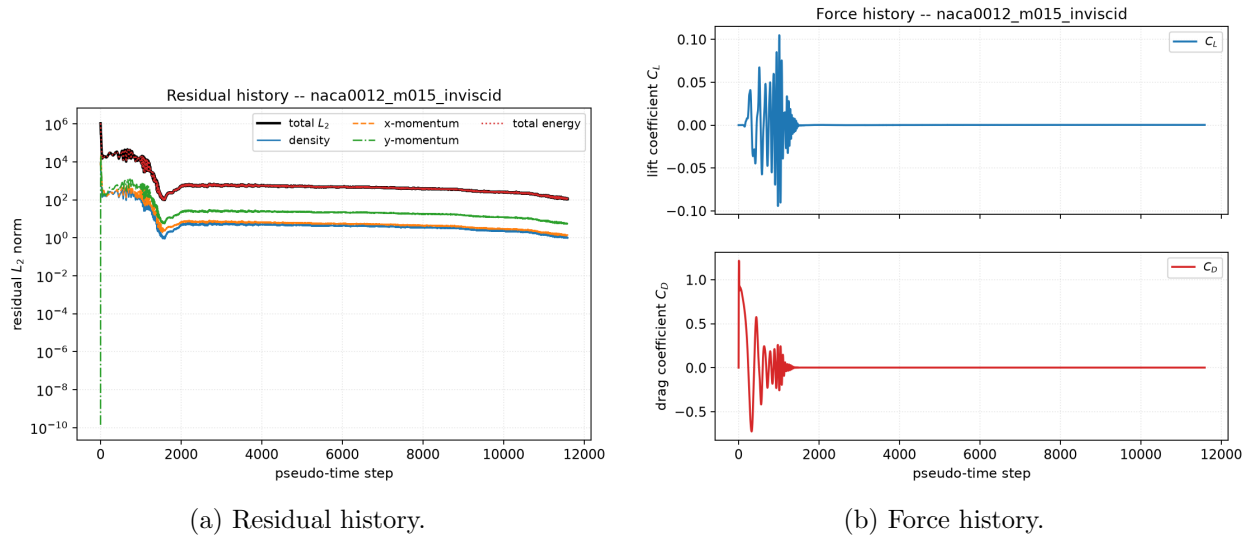


Figure 2: Convergence of `naca0012_m015_inviscid`: total and component-wise residual L_2 norms from `residuals.csv`, and lift and drag coefficient histories from `forces.csv`.

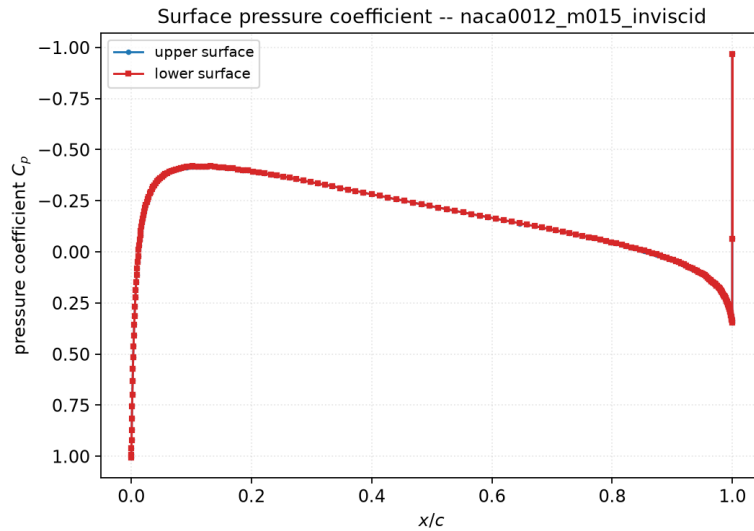


Figure 3: Surface pressure coefficient C_p against chordwise station for `naca0012_m015_inviscid`, from `surface.csv`, with the upper and lower surfaces separated by the sign of the wall normal and the ordinate inverted in the usual aerodynamic convention.

shock near $x/c \approx 0.55$. Roughly 20.6% of the near-body cells are supersonic. The pressure trace in Fig. 5a shows the corresponding plateau-then-jump signature on both surfaces, symmetric as it must be at zero incidence. The surface trace and the residual history are shown together in Fig. 5, the latter (Fig. 5b) reaching its four-order target in 6532 steps.

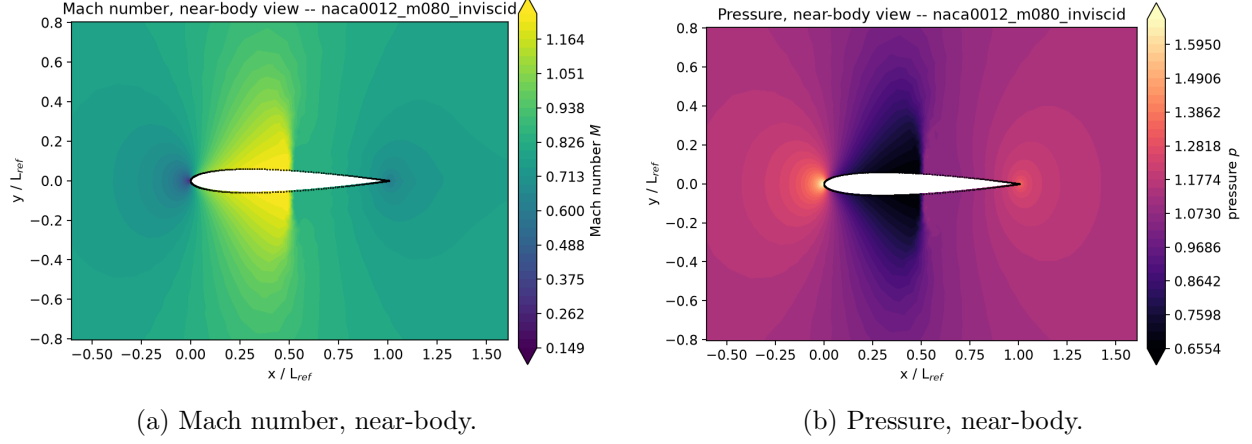


Figure 4: `naca0012_m080_inviscid`: the supersonic pocket over the forward half of the aerofoil and its terminating shock, in Mach number and pressure, from `field_final.vtu`. Colour ranges clipped to the 1st–99th percentile within the window.

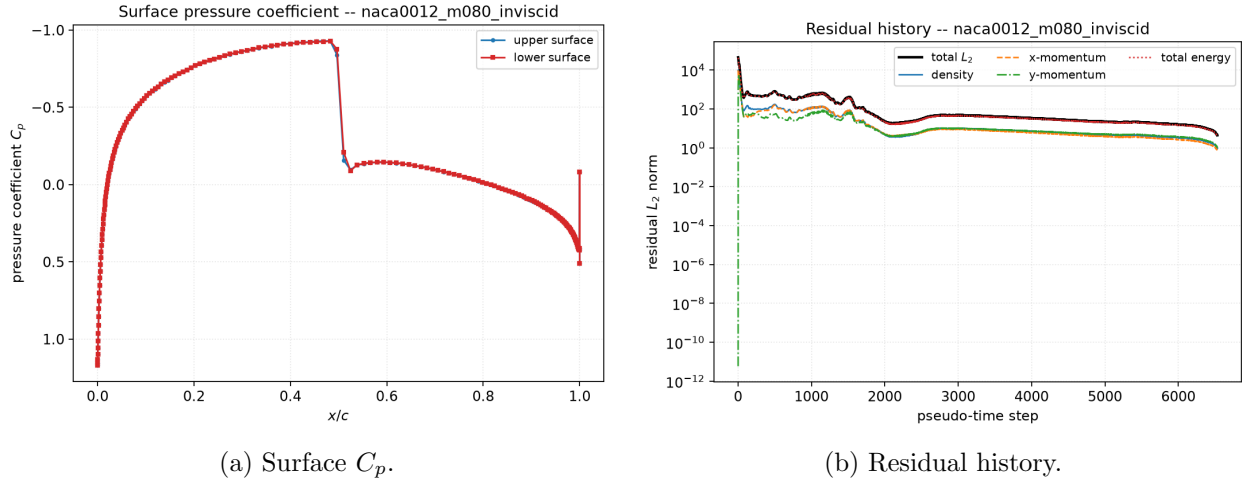


Figure 5: `naca0012_m080_inviscid`: surface pressure coefficient from `surface.csv` (ordinate inverted, upper and lower branches separated by wall-normal sign) and residual L_2 history from `residuals.csv`.

$M_\infty = 2.00$: **bow shock**. The supersonic case inverts the flow topology: the aerofoil now sits behind a detached bow shock, with a small subsonic region at the nose and supersonic flow almost everywhere else. Figure 6 shows this directly, in Mach number (Fig. 6a) and pressure (Fig. 6b): about 89.5% of the near-body cells are supersonic, the peak Mach number is 3.00 in the expansion around the shoulder, and the minimum is 3.2×10^{-2} at the stagnation point. Sampling above the upper surface gives $M = 0.74$ at the leading edge, crossing sonic by $x/c \approx 0.03$ and recovering to

$M \approx 2.1$ over the aft half. The drag is now dominated by wave drag: $C_D = 0.0880$ entirely from pressure, an order of magnitude above the subsonic value.

The supersonic cases are also where the limiter is most active, which caps the attainable residual floor: switching at the bow shock keeps a small residual component alive indefinitely. Their status in Table 6 is whatever the corrected criterion of Section 6.4 awarded, which requires settled forces in addition to the residual condition, and the measured numbers behind any plateau-branch label are audited in Table 9.

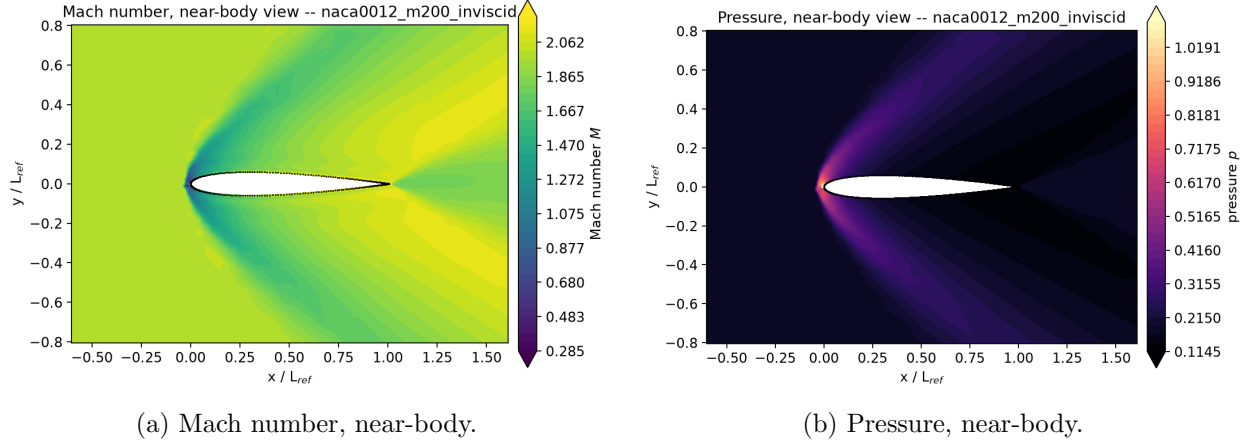


Figure 6: `naca0012_m200_inviscid`: bow-shock structure in Mach number and pressure from `field.final.vtu`, showing the subsonic nose pocket behind the detached shock and the supersonic expansion downstream. Colour ranges clipped to the 1st–99th percentile within the window.

10.3 NACA0012 laminar cases

The three laminar cases repeat the same Mach sweep at $Re = 5000$ with a no-slip adiabatic wall. Their defining feature is that skin friction, not pressure, sets the drag. At $M_\infty = 0.15$ the split is $C_{D,p} = 0.0182$ against $C_{D,v} = 0.0382$, so viscous drag is about twice the pressure drag and the total $C_D = 0.0564$ is roughly seventy times the inviscid value on the same mesh ($C_D = 8.2 \times 10^{-4}$). That ordering is the physically expected outcome for an attached laminar layer on a slender symmetric body at this Reynolds number, where form drag is small and friction dominates. The residual falls 5.17 orders in 3917 steps, the deepest reduction of the three laminar cases.

Figure 7 shows the $M_\infty = 0.15$ Mach field (Fig. 7a) and the wall distributions. The skin-friction curve in Fig. 7b peaks near the leading edge, where the boundary layer is thinnest and the wall-normal velocity gradient largest, and decays monotonically downstream — the classical laminar behaviour. Surface velocity is identically zero on these faces (Section 5), so the friction is computed from a genuine boundary condition rather than from an adjacent cell-centre value.

At $M_\infty = 0.80$ the laminar case adds shock–boundary-layer interaction to the transonic pocket, and it is the case that exposed the diagonal-splitting error described in Section 6.2, and Fig. 8 shows its near-body Mach field (Fig. 8a) beside the residual history that resulted (Fig. 8b). With the corrected operator the positivity limiting vanishes completely and the case now reaches its residual target outright: 4.23 orders in 7891 steps, with $C_D = 0.0802$ split $C_{D,p} = 0.0516$ against $C_{D,v} = 0.0286$. The relative C_D spread over the final 400 steps is 2.0×10^{-3} with drift 1.0×10^{-3} , so the forces are stationary as well as the residual small. Before that correction this case stalled at 2.60 orders and was the only one to stop on its step limit; the history of that stall is kept in

Section 6.2 because it is the clearest evidence that the diagonal splitting, not the mesh or the boundary layer, was the limit.

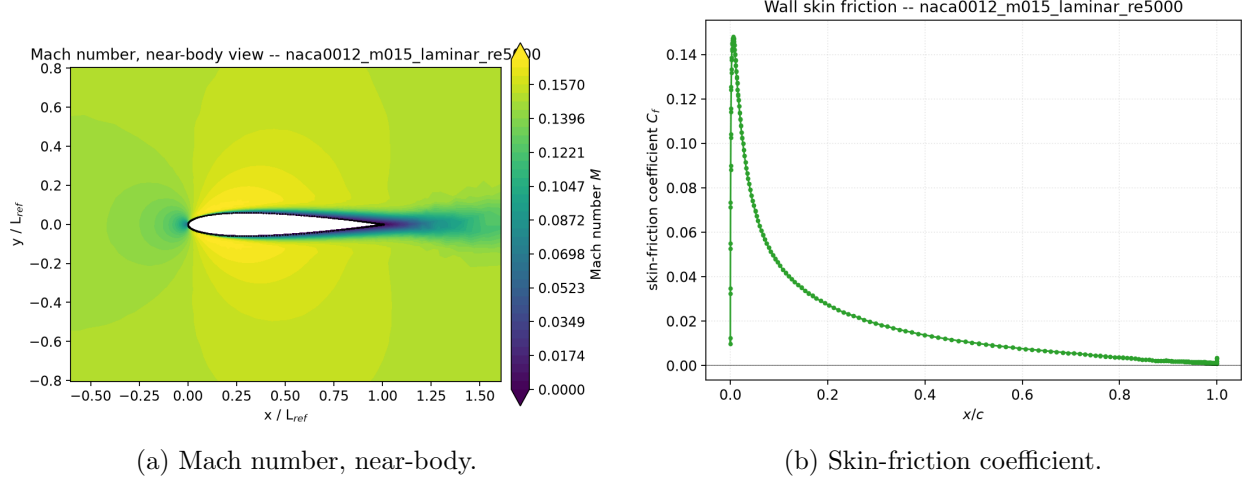


Figure 7: `naca0012_m015_laminar_re5000`: near-body Mach number from `field_final.vtu` and skin-friction coefficient C_f along the wall from `surface.csv`, computed from the tangential viscous traction only.

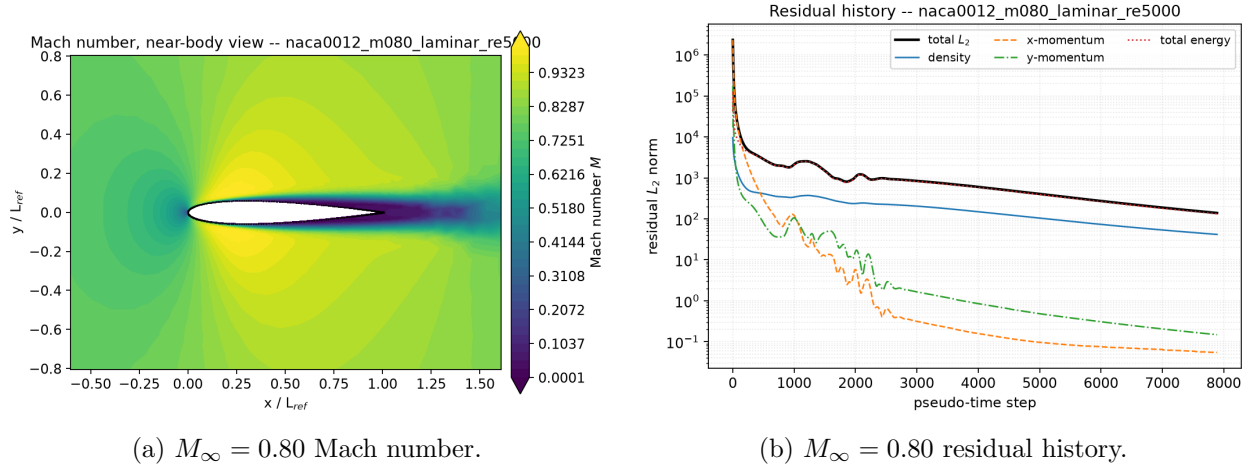


Figure 8: `naca0012_m080_laminar_re5000`: near-body Mach number from `field_final.vtu` and the residual L_2 history from `residuals.csv` after the diagonal-splitting correction of Section 6.2.

At $M_\infty = 2.00$ the laminar case combines the detached bow shock of its inviscid counterpart with a no-slip layer, and it is the most expensive steady case in the set: 25753 steps for 4.77 orders of residual reduction. The drag is $C_D = 0.1280$, split $C_{D,p} = 0.0883$ pressure against $C_{D,v} = 0.0397$ friction, so unlike the subsonic laminar case the wave drag now dominates the friction by better than two to one — the expected reordering once the bow shock is present. Lift is -7.1×10^{-4} , symmetric to the level the mesh supports. Figure 9 shows the near-body Mach field (Fig. 9a) and the wall skin-friction distribution (Fig. 9b), whose peak of $C_f = 0.0926$ sits just behind the nose where the post-shock boundary layer is thinnest; the small negative excursion to -0.0256 near the trailing edge marks the trailing-edge separation. The wall pressure coefficient spans $[-0.0707, 1.587]$, the upper

bound being the post-shock stagnation value that exceeds unity because the reference dynamic pressure is the freestream one.

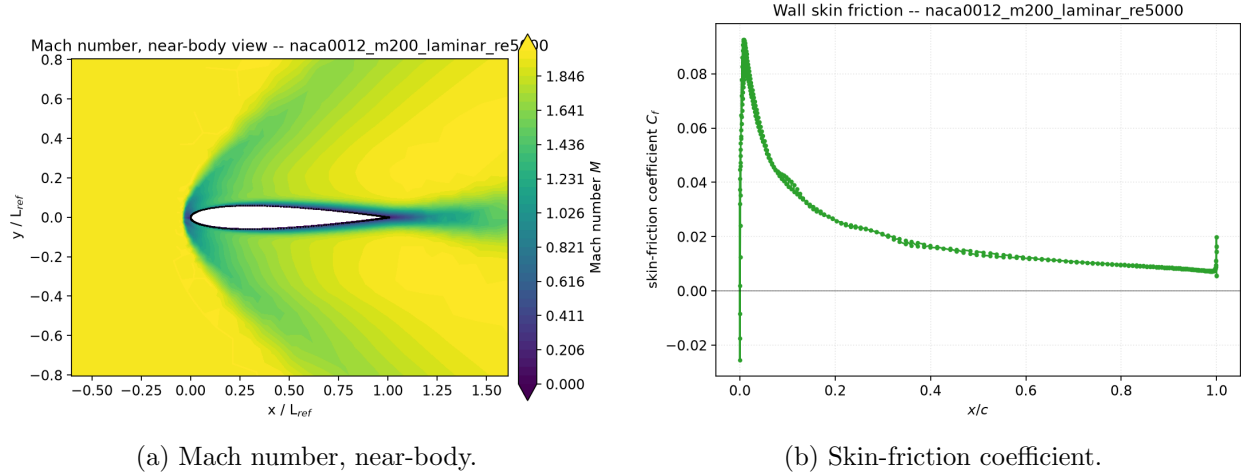


Figure 9: `naca0012_m200_laminar_re5000`: near-body Mach number from `field_final.vtu` and skin-friction coefficient C_f along the wall from `surface.csv`, from the tangential viscous traction only. Colour range clipped to the 1st–99th percentile within the window.

10.4 Cylinder at $Re = 20$

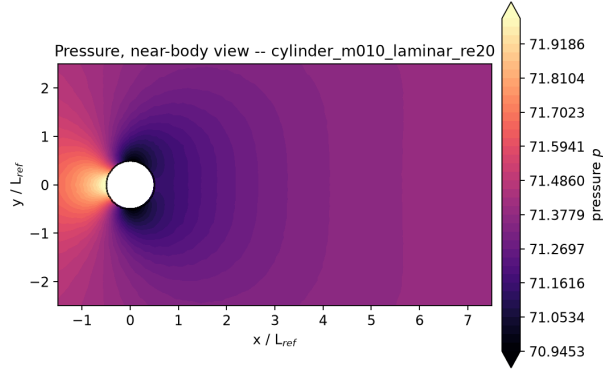
At $Re = 20$ the cylinder wake is steady and closed: two symmetric recirculating eddies behind the body with no shedding. This case is the strongest quantitative validation available in the benchmark, because the diameter-referenced drag coefficient of a circular cylinder at $Re = 20$ is well established in the literature at $C_D \approx 2.0$ – 2.1 .

The computed value is $C_D = 2.0181$, split as $C_{D,p} = 1.2228$ pressure and $C_{D,v} = 0.7953$ friction, with $C_L = 6.1 \times 10^{-4}$ — essentially zero, as symmetry demands. The agreement with the literature range is within about 1%, and the pressure/friction split near 60/40 is likewise the expected proportion at this Reynolds number. Figure 10 shows the pressure and Mach fields (Figs. 10a and 10b) and the wall distributions (Figs. 10c and 10d); the pressure recovers over the rear of the cylinder without reaching the front stagnation level, which is the signature of the separated wake.

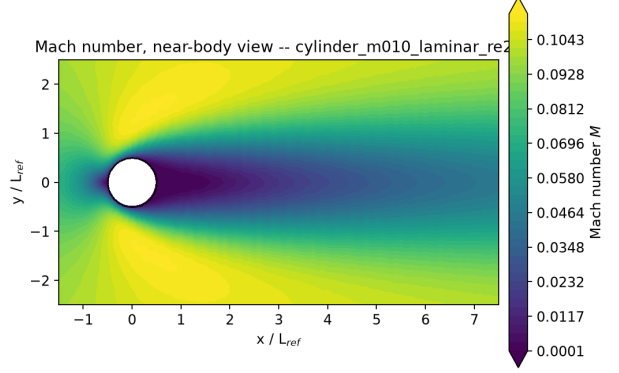
10.5 Cylinder at $Re = 200$: vortex street

At $Re = 200$ the steady symmetric wake is unstable and the flow settles into periodic vortex shedding. This case is computed with the dual-time BDF2 method of Section 6.5 at $\Delta t = 0.01$, and it is the only case whose correctness criterion is statistical periodicity rather than residual decay.

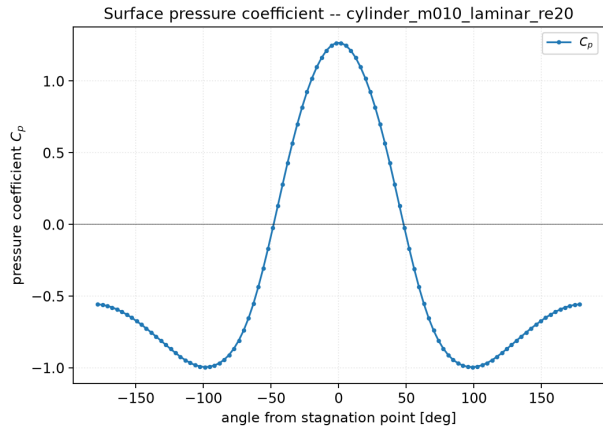
Shedding develops from the impulsively started field and is fully established well before $t = 40$. Statistics are therefore taken over $t \geq 40$, discarding the startup transient; the extraction is scripted in `tools/re200_stats.py` and its output is stored as `report/re200_stats.json` so the numbers quoted here are reproducible from the submitted force history. Over the post-transient window the results are:



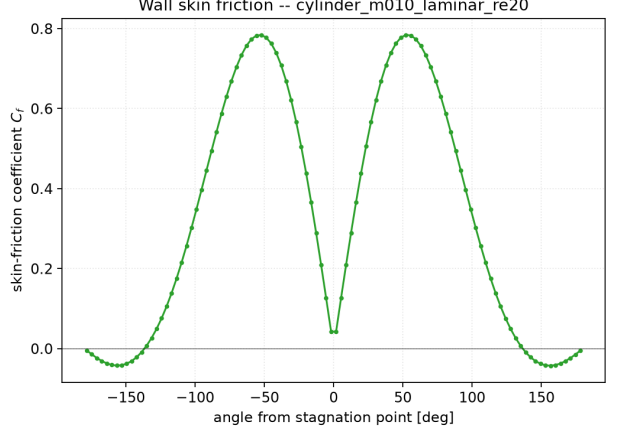
(a) Pressure, near-wake.



(b) Mach number, near-wake.



(c) Wall C_p against angle from stagnation.



(d) Wall C_f against angle from stagnation.

Figure 10: `cylinder_m010_laminar_re20`: near-wake pressure and Mach fields from `field.final.vtu`, and wall pressure and skin-friction coefficients from `surface.csv` plotted against the angle measured from the upstream stagnation point. Field colour ranges clipped to the 1st–99th percentile within the window.

Quantity	Computed	Reference
Mean C_D	1.236	≈ 1.3 (literature, $Re = 200$)
C_L amplitude	0.514	≈ 0.5
C_L mean	9.7×10^{-4}	0 by symmetry
Shedding period T	5.482	—
Strouhal number $St = fD/U$	0.182	≈ 0.19

The period is measured from upward zero crossings of C_L with linear interpolation between samples, averaged over 46 complete cycles; the cycle-to-cycle scatter is 0.0064 in period, i.e. 0.12%, which is itself the quantitative evidence that the signal is periodic rather than merely oscillatory. With $D = L_{\text{ref}} = 1$ and $U_\infty = 1$ the Strouhal number is simply $1/T = 0.182$, about 4% below the commonly quoted 0.19. The mean drag sits similarly below the literature figure. Both deficits are consistent with a wake that is slightly under-resolved: the cylinder mesh carries 10185 cells in total, which is modest for resolving the shear layers that set the shedding frequency, and coarse wake resolution characteristically lowers both St and C_D .

The mean lift is 9.7×10^{-4} against an amplitude of 0.514, i.e. the oscillation is symmetric about zero to within 0.2% of its amplitude, as it should be for a symmetric body.

Inner-solve statistics. The benchmark asks for the inner-iteration behaviour actually observed, not only the configured bounds, so `metadata.json` records it for every physical step of this run. The inner nonlinear solve took a mean of 161 iterations, with an observed minimum of 75 and maximum of 197 against the configured range of 5–1000. The inner target of 10^{-3} on the total transient residual was met on every one of the 30000 physical steps: `inner_target_misses` is 0 and `inner_target_converged_fraction` is 1.0, and the last physical step finished at a residual ratio of 9.76×10^{-4} . No physical step was accepted with an unconverged inner solve, which is the property that makes the time accuracy of the BDF2 march meaningful; a run that silently truncates its inner iterations degrades to something between BDF2 and a linearised step, with no way to tell from the force history alone.

Figure 11 shows the force history, in which the startup transient, the exponential growth of the wake instability, and the saturated periodic state are all visible. Figure 12 shows the wake structure, in vorticity (Fig. 12a) and velocity magnitude (Fig. 12b), from the post-transient field written at $t = 300$. The vorticity figure uses the documented clipped range $[-5, 5]$ recommended for this mesh: the raw vorticity extremes occur in the thin wall shear layers and are large enough to collapse the colour scale of the wake itself, so an automatic range would hide the very structure the figure exists to show. The wall distributions and the Mach and pressure fields for this case are collected in Appendix A.8.

11 Parallel Validation

Both required geometries were run at $np = 1, 2, 4$, and 8; Table 8 collects the results. Every run converged under the criterion of Section 6.4.

What this sweep establishes, and what it cannot, follows from the machine it was measured on. The benchmark asks that the parallelisation be correct: that the same problem solved on more ranks yield the same answer, and that the halo and reduction machinery carry no rank-dependent error. That is a question about solution consistency, and the sweep answers it. Wall-clock scaling is a separate question, and on this host it is not answerable, because the container is CPU-capped well below the rank counts the sweep uses. The cap is recorded rather than asserted, and it is

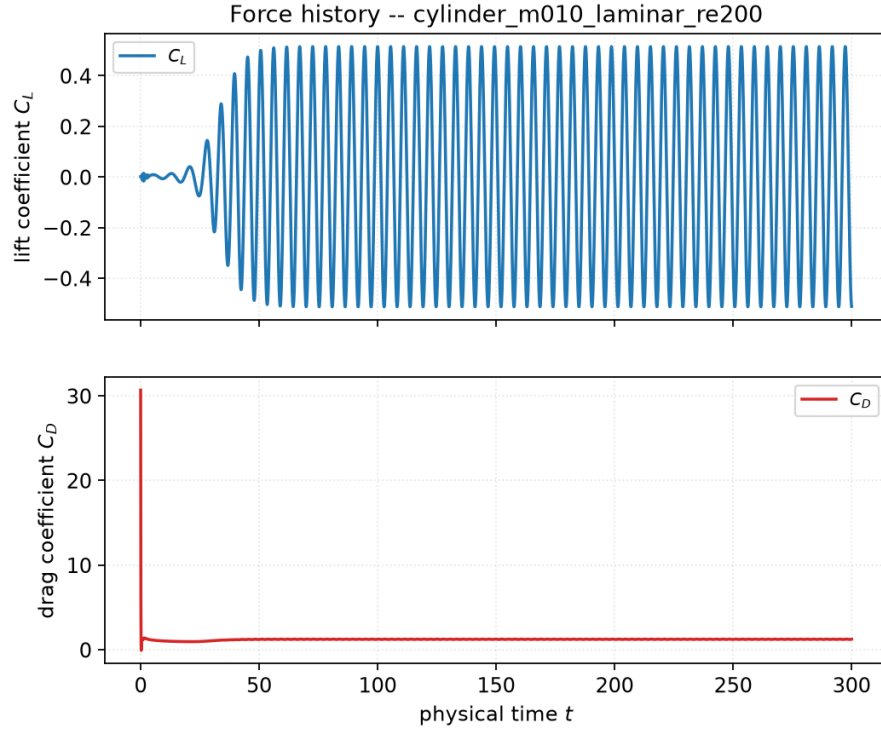


Figure 11: `cylinder_m010_laminar_re200`: lift and drag coefficient histories against *physical* time from `forces.csv`, showing the startup transient and the established periodic vortex shedding. Statistics in Section 10.5 use $t \geq 40$ only.

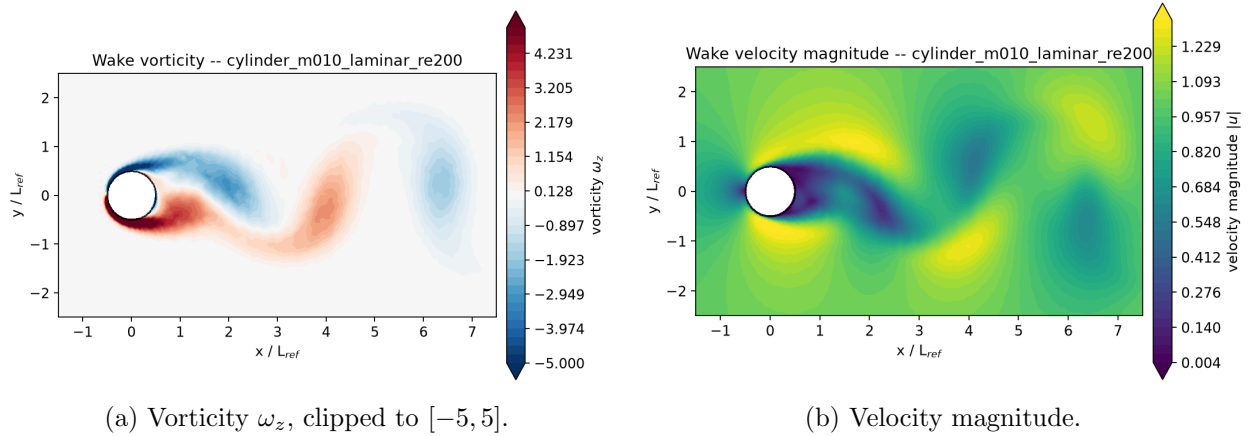


Figure 12: `cylinder_m010_laminar_re200`: post-transient wake structure from `field_final.vtu`. Vorticity is reconstructed from the cell-centred velocity by area-weighted nodal averaging followed by the curl on the triangulation, and is shown with the documented clipped range $[-5, 5]$; the velocity magnitude uses a 1st–99th percentile range.

Table 8: MPI rank-count study, read from `results/mpi_<case>_np<N>/`. The wall-time ratio is relative to the single-rank run of the same case, and is *not* a parallel speed-up: this host is capped at four CPUs, the runs shared it with other cases, and the column accordingly exceeds the CPU budget at $np = 4$ and falls at $np = 8$. It is reported because it is what was measured, not as an efficiency figure; see Section 11. The cylinder drag is an $O(1)$ quantity and agrees across rank counts to the precision shown; the NACA drag is itself of order 10^{-3} , so its variation sits at the convergence tolerance rather than indicating a rank-dependent discrepancy.

Ranks	Steps	Orders	Wall (s)	Wall ratio	C_D
<i>naca0012_m015_inviscid</i>					
1	11726	4.003	8374.9	1.00	0.000822
2	11731	4.004	2732.7	3.06	0.000821
4	11591	4.001	891.2	9.40	0.000822
8	11774	4.004	1011.2	8.28	0.000817
<i>cylinder_m010_laminar_re20</i>					
1	2469	5.593	279.8	1.00	2.017874
2	2489	5.583	236.2	1.18	2.017984
4	2529	5.582	28.2	9.94	2.018071
8	2605	5.542	46.8	5.98	2.018504

stated below before any timing is discussed, so the timings are read in the light of the constraint they were measured under.

The CPU allowance of the measurement host. The operating system advertises 64 cores, and that figure is misleading here, because the cgroup that owns this container caps the CPU time the container may consume. Its `cpu.max` value is 400000 100000, a quota and a period in microseconds, granting 4 CPUs of runnable time per scheduling period of 100000 microseconds, however many cores are visible. The kernel also reports how often that ceiling was reached. `tools/cpu.budget.py` samples `cpu.max` and `cpu.stat` and stores them as a citable artifact; at the sampling recorded there, 287148 of 571074 scheduling periods were throttled, about 50%, withholding some 21515 seconds of runnable time. These are kernel counters, not an inference from observed slowness.

The consequence is arithmetic. At $np = 8$ the run requests 2.0 ranks per available CPU, so eight processes share 4 CPUs of budget and each advances at roughly half the rate a dedicated core would give it, before any communication cost is counted. A speed-up curve measured under that constraint reports the quota rather than the solver, and no parallel efficiency is claimed from it anywhere in this report.

Solution consistency. The cylinder is the decisive test, because its drag is an $O(1)$ quantity that the scheme resolves well, so a rank-dependent error would appear as a change in a number that is large enough to read unambiguously. Across all four rank counts the converged drag is 2.017874, 2.017984, 2.018071, and 2.018504, a relative spread of 3.1×10^{-4} . The comparison is stringent because the partition is not a passive relabelling: it changes the LU-SGS relaxation, the halo pattern, the reduction tree, and the order in which floating-point contributions accumulate. A converged load that survives all four is evidence that none of those mechanisms introduces a rank-dependent error.

The NACA drag, 0.000822, 0.000821, 0.000822, and 0.000817, has a larger relative variation, and

it has to be read differently. At $\alpha = 0$ a symmetric section in inviscid flow has no physical drag: the exact answer is zero, so the computed value is not a physical quantity resolved to some accuracy but discretisation error alone, here of order 10^{-3} . Variation between rank counts is therefore variation *within* the numerical noise floor, and the four runs agree on the substantive statement, that the drag is $O(10^{-3})$ and indistinguishable from zero. A relative difference between two values that are both numerically zero carries no information, which is why the steadiness measure of Eq. (18) carries an absolute scale floor alongside its relative term.

Iteration counts. Step counts that track each other across rank counts are evidence that the halo exchange and sweep ordering are correct, since stale, misordered or misaddressed halo data would change the relaxation and with it the number of steps needed to reach the residual target. The cylinder spans 2469 to 2605 steps across the sweep, and the drift is in the expected direction: LU-SGS is a rank-local relaxation, with ghost-cell increments frozen within a sweep (Section 6.2), so more partitions mean more frozen ghost increments per sweep and each sweep is a slightly weaker approximation to a global one. A rank-independent step count would in fact be the surprising outcome for this solver; a mild increase with rank count is what the algorithm implies.

Wall times, and why they are not a scaling result. The measured wall times are 8374.9, 2732.7, 891.2, and 1011.2 seconds for the aerofoil and 279.8, 236.2, 28.2, and 46.8 seconds for the cylinder, at $np = 1, 2, 4$ and 8 respectively. They are reported because the benchmark asks what the runs cost, and they are not converted into an efficiency because four independent effects contribute to them and only one is a property of the parallelisation.

The first is the CPU quota recorded above. Beyond 4 ranks the additional processes contend for a fixed budget, so the $np = 8$ column cannot improve on $np = 4$ by more than the small serial-fraction margin, and will generally be worse: eight ranks under a four-CPU cap add context switching, MPI progress threads and synchronisation delay at every exchange, while the total compute throughput available to the job is unchanged. Inverted scaling at the top of the sweep is the expected result, not an anomaly.

The second is that the runs do not perform equal work. The step count adapts per partition, as above, so a higher-rank run that needs more steps is solving a longer march, and dividing two wall times compares different amounts of computation.

The third is problem size. The meshes carry 20816 and 10185 cells, so at $np = 8$ each rank owns roughly 2600 and 1270 cells. Per-rank work per step is then small enough that the halo exchange and the global reductions for residual and force norms are no longer negligible beside it, which is the ordinary strong-scaling limit of a fixed problem divided too finely.

The fourth is that the sweep did not have the host to itself: the rank-sweep runs were executed alongside other production cases from the same regeneration script, so each timing carries contention from whatever else held the quota at that moment. This is visible in the sweep itself, where wall times at neighbouring rank counts differ by more than any model of the solver would predict, and it is the reason no speed-up figure from this table is quoted as a property of the code. The timings bound the cost of the runs on this host; they do not measure parallel efficiency, and establishing that would require an uncapped, unshared machine.

Communication volume is small in proportion to work: on the busiest NACA rank the halo is 201 cells against 2613 owned (Table 4), a 7.7% surface-to-volume overhead, so a state exchange moves 201×4 doubles against a residual assembly that visits thousands of faces. This ratio is the intended consequence of a neighbour-scoped pattern over a partition with a low METIS edge cut, and it is what makes the communication term small enough that the effects above, rather than

message volume, dominate the measured times.

Telemetry carried in the per-step reductions. Part of the per-step cost is reporting rather than solving, and the distinction differs between the two drivers. The transient driver performs one additional small reduction per physical step, carrying the positivity-fallback and limiter counts added late in this work for the diagnostics of Section 8. There the counts are used only for reporting: they are summed, recorded and logged, and feed no solver state. The reduction is deliberately placed outside the inner iteration and both counters travel in one collective, so the addition is a single round trip per physical step rather than one per sweep. At high rank counts, where a reduction is a synchronisation point and the per-rank work between reductions is small, a little of the per-step cost is therefore attributable to telemetry rather than to solving.

In the steady driver, which is the one the runs of this section use, the corresponding counts are not purely diagnostic and the report should not describe them as such. The summed positivity-limiting count drives the CFL safety factor of Section 6.3, and the summed nonphysical-cell count enters the final convergence verdict, so those reductions are part of the algorithm and their values affect the march. What the late change did there was merge a new counter into a reduction the step already performed, leaving the number of collectives per step unchanged.

In both drivers the counts are small non-negative integers carried in a double-precision sum, which represents them exactly, so merging or adding a counter changes no reduced value. That this leaves the solution unchanged was checked rather than assumed: the pre-change commit and the current source were each built and run on the same case for the same bounded horizon, and `forces.csv`, `residuals.csv`, `surface.csv` and the full double-precision `restart_final.bin` agree byte for byte (`docs/bit_identity_check*.txt`), with the build history and the affected commits recorded in `docs/binary_provenance.md`. The scope of that evidence is stated in the check itself and is worth repeating: the comparisons are single-rank, and at one rank every reduction in the changed code is bypassed, so they establish that the changes do not perturb a computed state without exercising the multi-rank reduction path.

12 Visualisation Approach

All field figures are filled contour plots drawn on the actual computational cells. Quadrilaterals are split into two triangles and the mixed mesh is handed to a single `matplotlib` triangulation, so the plotted field follows the mesh rather than an interpolation grid, and no figure in this report is a scatter of points.

Two presentation choices deserve a note. Colour ranges are clipped to the 1st–99th percentile of the data *within the plotted window*, and the clipping is stated in the figure manifest caption for every affected figure. Window-local clipping matters more than it might appear: these meshes extend 80 body lengths to the farfield, so a range computed over the whole domain is set by uniform freestream cells and flattens every near-body feature. For the same reason each field is shown in a bounded window derived from the wall bounding box in `surface.csv` — expanded symmetrically for the slender aerofoil and stretched downstream for the bluff cylinder — rather than as a whole-domain view in which the body occupies a few pixels. The vorticity figure is the one case with a fixed rather than percentile range, using the documented $[-5, 5]$ band.

13 Extensibility

The benchmark asks that the code leave clear programming space for 3-D, general equations of state, RANS turbulence and multi-species flow. The honest answer is that three of those four are localised changes and one is not, and it is more useful to say which is which than to claim uniform readiness.

State-vector width: parameterised. `kNumVars = 2 + kDim` is a load-bearing constant, not decoration. The residual loops, the LU-SGS block solve, the halo payload width and the restart record are all written against it, so adding a turbulence variable or extra species widens the state without touching the transport machinery. The one caveat is that energy is still indexed as a literal `u[3]` at a handful of sites outside the equation of state, which would move to `u[4]` in 3-D.

Equation of state: behind an interface. `PerfectGas` owns every thermodynamic conversion, and the flux routines ask it for pressure, temperature and sound speed rather than computing them. A real-gas or tabulated EOS replaces that class. Nothing in the Riemann solvers or the viscous flux assumes the ideal relation except through those calls.

RANS: a hook exists, the model does not. Every RANS model is defined by its source terms, and a flux-divergence residual with no volumetric term has nowhere to put them. `evaluate` therefore calls `accumulateSources` explicitly after the interior and boundary face loops. It is empty for this equation set — the compressible Navier–Stokes equations for a perfect gas have no volumetric production — and exists so a model can be added at a defined call site instead of by editing the residual loop. Turbulent viscosity is already a caller-supplied scalar into the viscous flux, fetched at two sites, so an eddy-viscosity model needs a μ_t field and those two lookups. A stiff destruction term would additionally want its Jacobian on the implicit diagonal.

Three dimensions: not parameterised. This is the one that would require real work, and the code should not pretend otherwise. `Vec2` has two components and appears in 25 files; `cross()` returns a scalar, which is inherently two-dimensional and drives cell area, centroid, winding and the moment; a face is hard-wired to exactly two nodes, with the face-match hash packing two vertex ids into one 64-bit word; and `MeshCell::vertices` is a four-element array, which no hexahedron fits. A 3-D extension means a new geometry and face-connectivity layer — a face becomes a polygon — plus a `Vec2` to `Vec3` migration. What survives that change is everything above the geometry: the flux functions, the limiter, the implicit solver, the partitioner interface and the halo pattern are all written against faces and states rather than against coordinates. The header comment in `types.hpp` says this explicitly rather than advertising dimension independence the geometry does not have.

14 Limitations and Failure Analysis

Cost of running at the supplied parameters. The $Re = 200$ transient is the expensive case, and it is expensive because it is run at the supplied inner relaxation pseudo-CFL of 1.0 (Section 6.6). At that value the inner fixed-point iteration needs roughly 104 sweeps per physical step against roughly 30 at a pseudo-CFL of 10, so the 30000-step march costs several times what it otherwise would. An earlier version of this solver took the cheaper route behind a hard-coded constant; that is described in Section 6.6 and was removed. The remaining honest limitation is that no cheaper

equivalent has been *demonstrated* at the supplied setting — the argument that the relaxation parameter cancels at the BDF2 root is sound, but the submission now simply pays the cost rather than relying on that argument.

Approximate implicit operator. A note on the CFL safety controller of Section 6.3: it is a heuristic, and its first tuning was actively harmful, producing a self-inflicted residual plateau on the $M_\infty = 0.80$ laminar case. The retuned version is better behaved, but a controller of this kind is a workaround for the absence of a viscous Jacobian contribution rather than a substitute for one. A robust implementation would make the aggressive step safe rather than detecting and retreating from it.

Three approximations sit in the implicit operator, all of which affect the path to the steady state but not the steady state itself:

- off-diagonal blocks use a scalar spectral radius rather than the full neighbour Jacobian, so the LU-SGS sweep is a genuine approximation to a Newton step;
- the diagonal block carries no viscous Jacobian contribution — the viscous term enters only through the spectral radius in Eq. (17) and the diagonal weighting of Section 6.2. A proper thin-shear-layer Jacobian would likely improve the laminar convergence rate, particularly at $M_\infty = 0.80$;
- ghost-cell increments $\Delta \mathbf{U}$ are frozen within a sweep, so the Gauss–Seidel coupling stops at partition boundaries. This is what makes the sweep rank-local, and it is the direct cause of the mild rank-dependence of the convergence path discussed in Section 11.

Cases terminating on a plateau rather than on the residual target. Some steady cases stop for a reason other than reaching their nominal residual target, and each is reported with its actual measured value in Table 6 rather than relabelled. Cases qualifying through the plateau branch of Section 6.4 are marked there with a dagger. The audit trail for each is its `notes` field in `run_status.json`, which records the measured residual orders, relative C_D spread, and drift that the plateau test used; Table 9 reproduces those numbers.

Table 9: Audit of every case labelled **converged** through the plateau branch of Section 6.4. Values are parsed from the `notes` field written by the solver into `run_status.json`; the plateau test requires residual orders ≥ 2.5 , relative C_D spread $< 10^{-2}$, and drift $< 10^{-3}$.

Case	Orders	C_D spread	C_D drift	Passes test
NACA0012 <i>M2.00</i> inviscid	2.55	0.0069	2.190×10^{-5}	yes

The plateau branch is not a softer standard. It relaxes only the residual requirement, from the case target to 2.5 orders; the force-steadiness test of Eq. (18) is identical in both branches, so a plateau-labelled case has demonstrated exactly the same settled-load evidence as one that reached its residual target. What the branch concedes is that on some meshes the attainable residual floor is set by mechanisms other than the solution error — limiter switching at a shock, or the CFL safety controller of Section 6.3 engaging on high-aspect-ratio near-wall cells — and that continuing to march against such a floor changes the loads not at all.

Table 9 is generated directly from the `notes` fields, so it reflects whatever the final production sweep produced rather than any earlier run. Where it reports that no case required the plateau branch, every steady case reached its full residual target with settled forces.

Wake resolution at $Re = 200$. The computed Strouhal number 0.182 and mean drag 1.236 both sit a few percent below the accepted values (≈ 0.19 and ≈ 1.3). The cylinder mesh has 10185 cells, which is coarse for the shear layers that set the shedding frequency, and under-resolution lowers both quantities in the direction observed. This is a mesh-resolution limitation, not a time-integration one: the shedding is cleanly periodic (0.12% period scatter) and the time step is small.

Nondimensionalisation caveat. Cylinder force coefficients are referenced to $L_{\text{ref}} = 1$, which equals the cylinder diameter for these meshes, so the reported C_D is diameter-referenced and directly comparable with the literature. Readers comparing against radius-referenced conventions must rescale.

Other known limits. The gradient stencil uses face neighbours only, which is the minimum viable stencil and is less robust on highly skewed cells than a vertex-neighbour stencil. The Sutherland viscosity law is implemented but unused, since all supplied cases specify constant viscosity. There is no turbulence model: the laminar cases are solved as laminar, which is what the benchmark specifies but is not physical at $Re = 5000$ over a real aerofoil. Finally, the field output is ASCII VTK, which is convenient, human-readable and portable, but substantially larger than a binary equivalent would be: the NACA field files are 4.8MB each for 20816 cells, and the base64/raw binary VTU encodings would cut that severalfold.

A defect found and fixed during this work. One bug is worth recording because it was silent. The field writer initially emitted cell connectivity using rank-local vertex indices where global ids were required. At $np = 1$ the two numberings are permutations of each other, so every point was referenced and the output looked plausible while the topology was scrambled; at $np = 8$ only 2326 of 15682 points were referenced. It was caught not by inspection but by a geometric invariant — summed cell area against the mesh bounding box — which is now checked for every field file (Section 8). The lesson generalises: a rendering that looks reasonable is weak evidence, and cheap geometric invariants are strong evidence.

15 Conclusions

`ufvm2d` solves all eight required cases with a single executable and a single code path, driven entirely by the case JSON with no case-specific branching anywhere in the solver. All seven steady cases reach the residual reduction target their own case file specifies — which is three orders for the two supersonic cases and four or five for the rest, so “three orders” for those two is target attainment rather than a shortfall — and each also passes the force-steadiness test of Section 6.4. The unsteady cylinder reaches clean statistical periodicity. Quantitatively the strongest external validation is the $Re = 20$ cylinder drag, $C_D = 2.0181$ against a literature range of 2.0–2.1; the transonic and supersonic aerofoil cases reproduce the expected shock structure; and the $Re = 200$ shedding frequency is within a few percent of the accepted Strouhal number on a deliberately modest mesh.

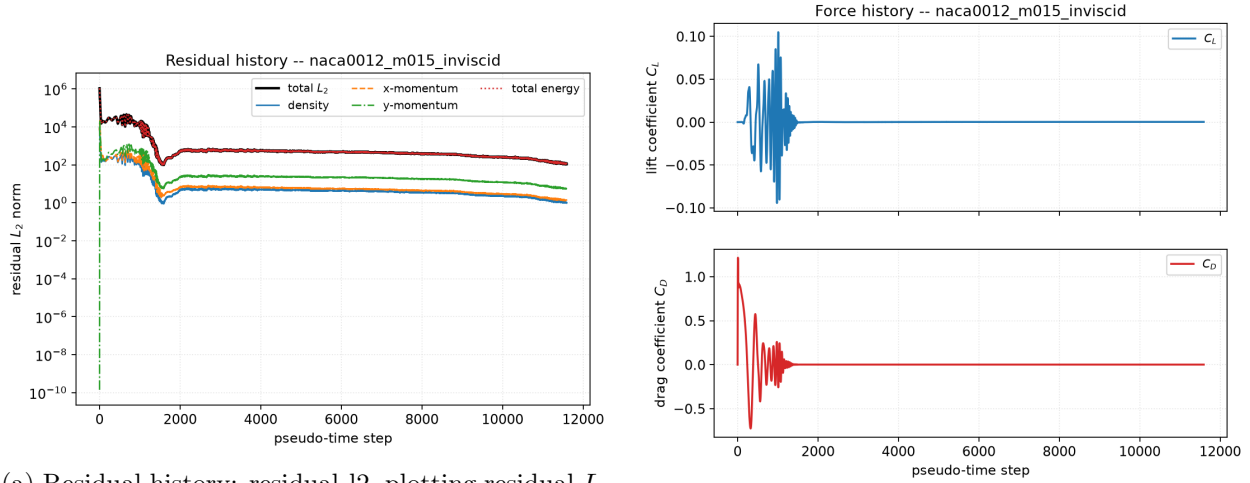
The parallel implementation meets the structural requirements of the benchmark rather than merely appearing to: the global mesh is released after partitioning, no rank holds full state, and halo data moves only through neighbour-scoped non-blocking exchanges. Where results fall short — a slightly low Strouhal number on a coarse wake mesh — this report says so explicitly rather than presenting them as successes.

A Complete Figure Atlas

This appendix reproduces every figure listed in `report/figure_manifest.csv`, grouped by case. Figures discussed in Section 10 appear there as well; collecting them here guarantees that the set of figures in this report and the set mapped by the manifest are the same set, and that each one carries a caption naming the variable plotted and the data file it was drawn from.

A.1 NACA0012, $M_\infty = 0.15$, inviscid

Figures 13 to 16 reproduce every figure produced for `naca0012_m015_inviscid` in the order they appear in `figure_manifest.csv`, which records the source data file and the plotted variable for each one.



(a) Residual history: residual_l2, plotting residual L_2 norm, from `residuals.csv`. Residual L_2 norm and the four conserved-component norms versus pseudo-time step for `naca0012_m015_inviscid`; total reduction 4.00 orders of magnitude.

(b) Force history: cl_cd, plotting C_L and C_D , from `forces.csv`. Lift and drag coefficient history versus pseudo-time step for `naca0012_m015_inviscid`; final $C_L = 0.000256146$, $C_D = 0.000822021$.

Figure 13: NACA0012, $M_\infty = 0.15$, inviscid: figure set 1 of 4.

A.2 NACA0012, $M_\infty = 0.80$, inviscid

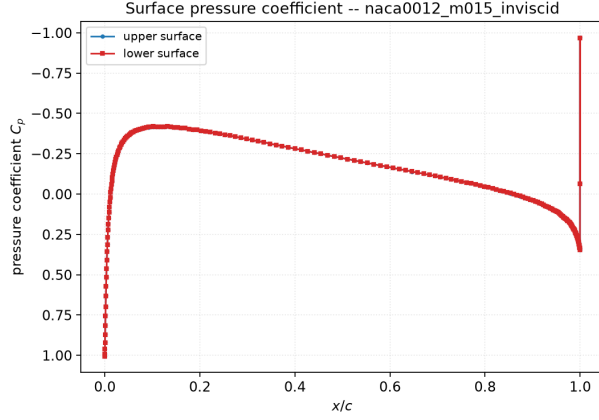
Figures 17 to 20 reproduce every figure produced for `naca0012_m080_inviscid` in the order they appear in `figure_manifest.csv`, which records the source data file and the plotted variable for each one.

A.3 NACA0012, $M_\infty = 2.00$, inviscid

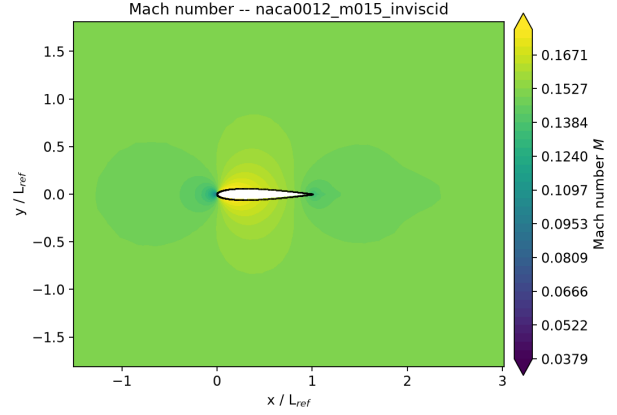
Figures 21 to 24 reproduce every figure produced for `naca0012_m200_inviscid` in the order they appear in `figure_manifest.csv`, which records the source data file and the plotted variable for each one.

A.4 NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$

Figures 25 to 28 reproduce every figure produced for `naca0012_m015_laminar_re5000` in the order they appear in `figure_manifest.csv`, which records the source data file and the plotted variable

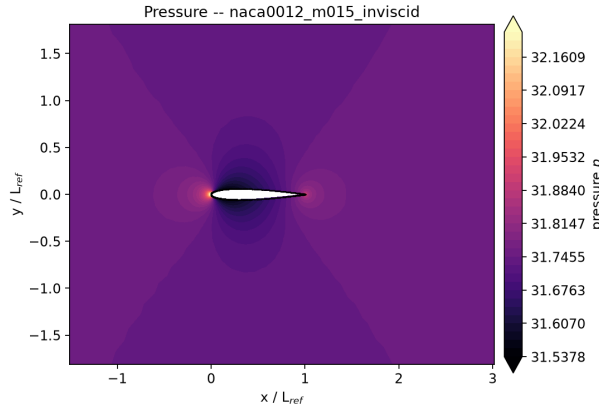


(a) Wall pressure coefficient: C_p , plotting C_p , from `surface.csv`. Wall pressure coefficient for `naca0012.m015.inviscid` versus chordwise station with the ordinate inverted; 404 surface samples, C_p range $[-0.972, 1.006]$.

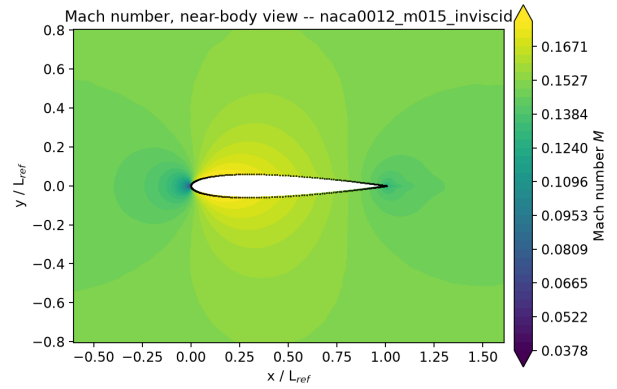


(b) Field contours, full window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.03786, 0.1779]$ of the full range $[0.006561, 0.1781]$.

Figure 14: NACA0012, $M_\infty = 0.15$, inviscid: figure set 2 of 4.

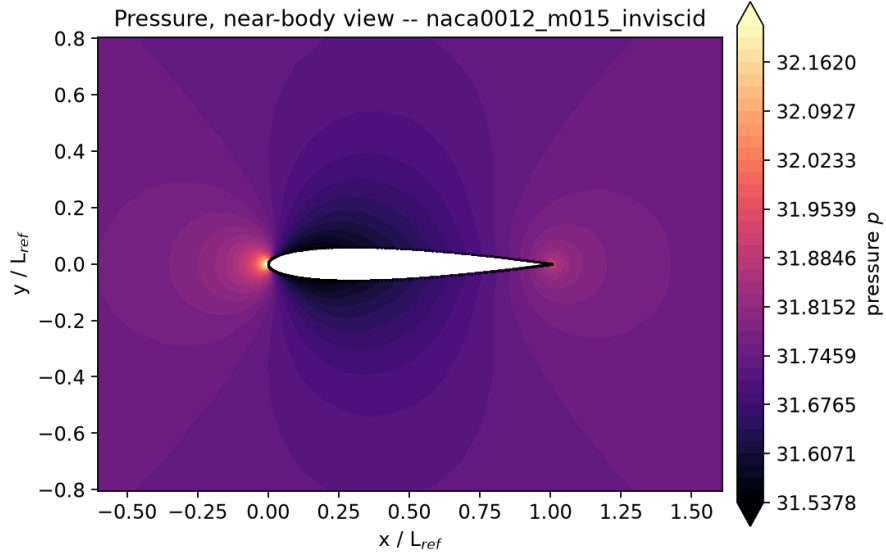


(a) Field contours, full window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[31.54, 32.21]$ of the full range $[31.41, 32.25]$.



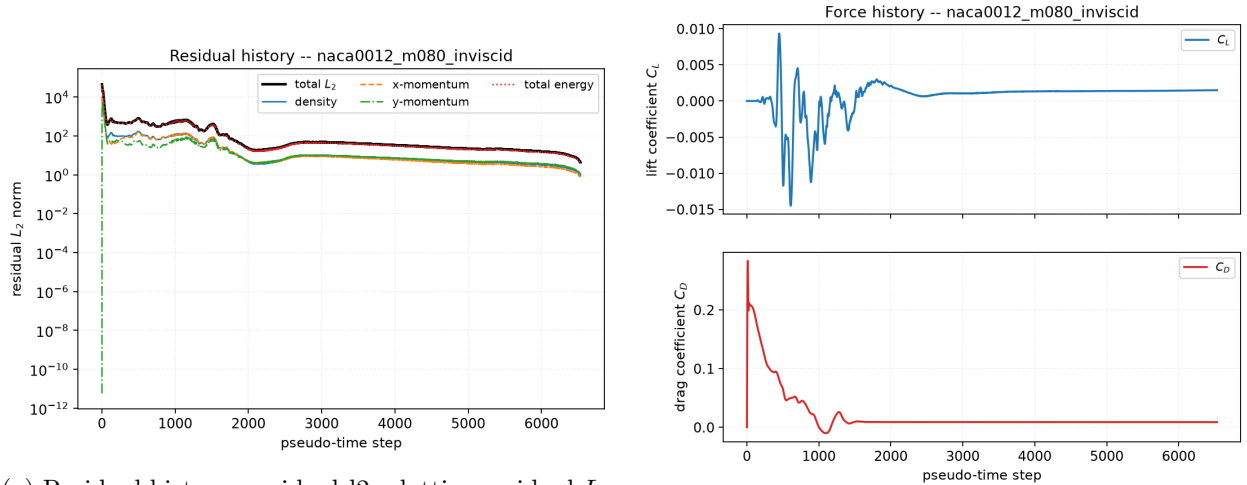
(b) Field contours, near-body window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.03778, 0.1779]$ of the full range $[0.006561, 0.1781]$.

Figure 15: NACA0012, $M_\infty = 0.15$, inviscid: figure set 3 of 4.



(a) Field contours, near-body window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile [31.54, 32.21] of the full range [31.41, 32.25].

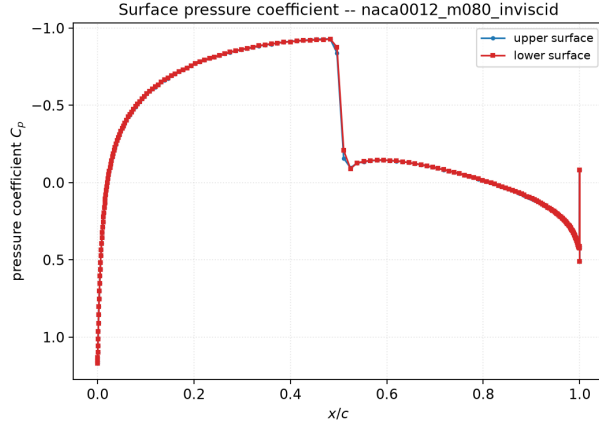
Figure 16: NACA0012, $M_\infty = 0.15$, inviscid: figure set 4 of 4.



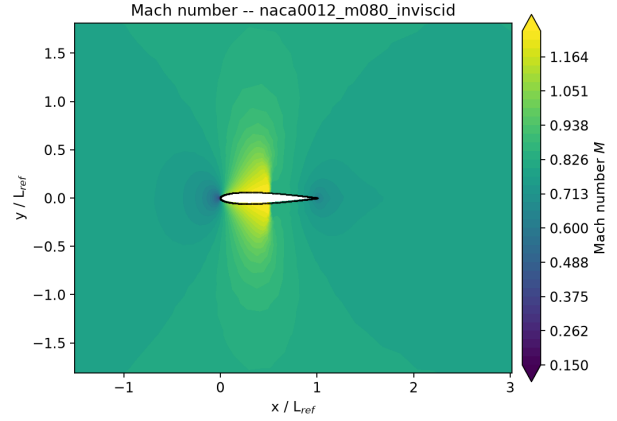
(a) Residual history: residual_l2, plotting residual L_2 norm, from `residuals.csv`. Residual L_2 norm and the four conserved-component norms versus pseudo-time step for `naca0012_m080_inviscid`; total reduction 4.00 orders of magnitude.

(b) Force history: `cl_cd`, plotting C_L and C_D , from `forces.csv`. Lift and drag coefficient history versus pseudo-time step for `naca0012_m080_inviscid`; final $C_L = 0.00149483$, $C_D = 0.00872514$.

Figure 17: NACA0012, $M_\infty = 0.80$, inviscid: figure set 1 of 4.

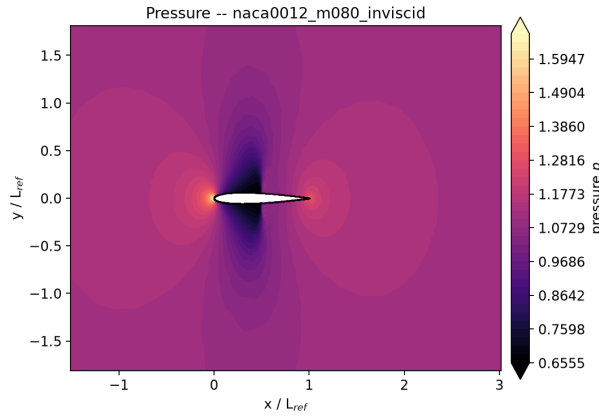


(a) Wall pressure coefficient: C_p , plotting C_p , from `surface.csv`. Wall pressure coefficient for `naca0012_m080_inviscid` versus chordwise station with the ordinate inverted; 404 surface samples, C_p range $[-0.9289, 1.169]$.

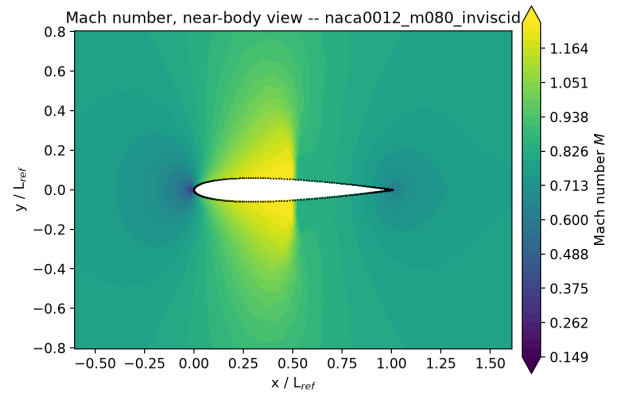


(b) Field contours, full window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.1496, 1.248]$ of the full range $[0.02503, 1.256]$.

Figure 18: NACA0012, $M_\infty = 0.80$, inviscid: figure set 2 of 4.

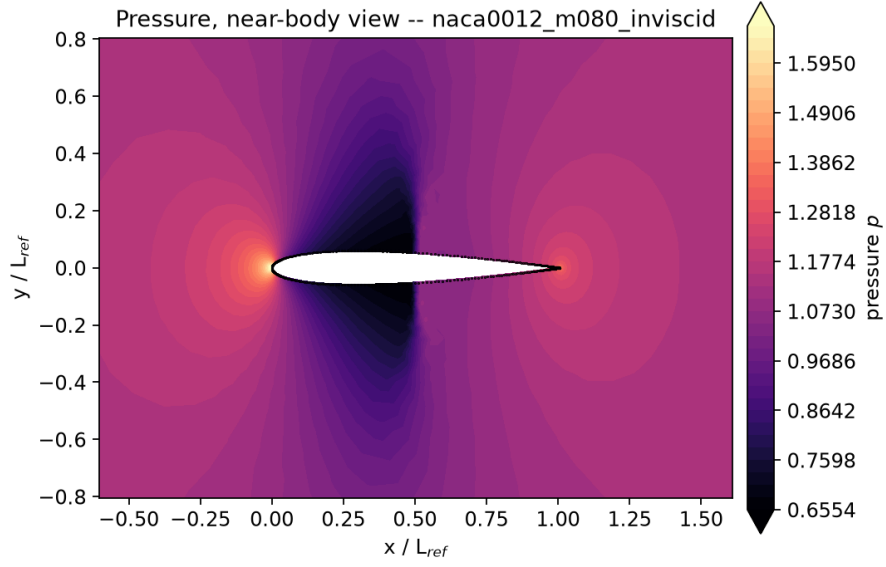


(a) Field contours, full window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.6555, 1.673]$ of the full range $[0.6517, 1.701]$.



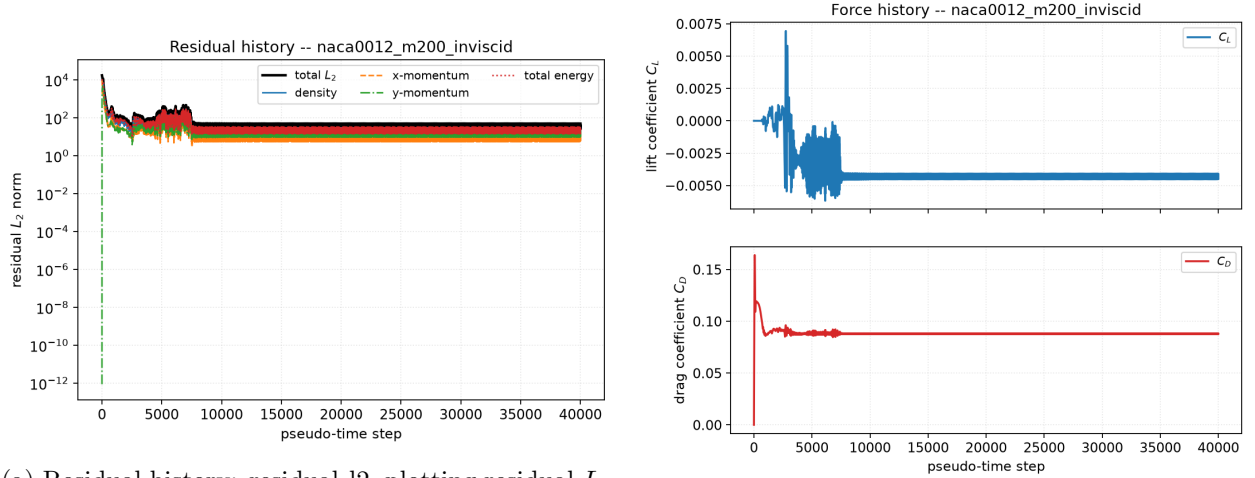
(b) Field contours, near-body window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.1495, 1.248]$ of the full range $[0.02503, 1.256]$.

Figure 19: NACA0012, $M_\infty = 0.80$, inviscid: figure set 3 of 4.



(a) Field contours, near-body window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile [0.6554, 1.673] of the full range [0.6517, 1.701].

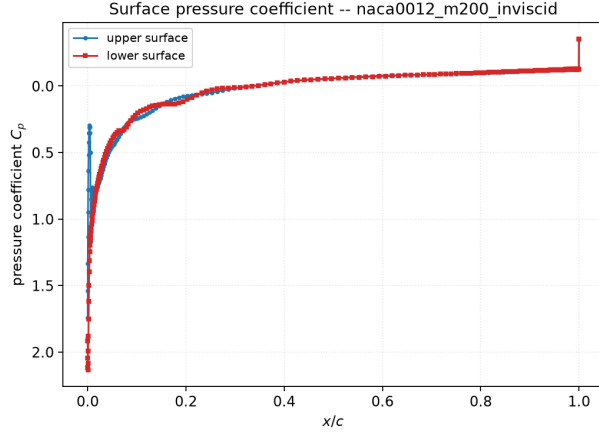
Figure 20: NACA0012, $M_\infty = 0.80$, inviscid: figure set 4 of 4.



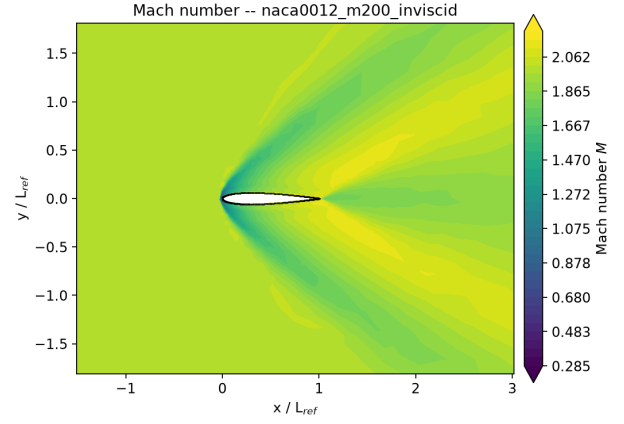
(a) Residual history: residual_l2, plotting residual L_2 norm, from `residuals.csv`. Residual L_2 norm and the four conserved-component norms versus pseudo-time step for `naca0012_m200_inviscid`; total reduction 2.55 orders of magnitude.

(b) Force history: `cl_cd`, plotting C_L and C_D , from `forces.csv`. Lift and drag coefficient history versus pseudo-time step for `naca0012_m200_inviscid`; final $C_L = -0.00447707$, $C_D = 0.0879675$.

Figure 21: NACA0012, $M_\infty = 2.00$, inviscid: figure set 1 of 4.

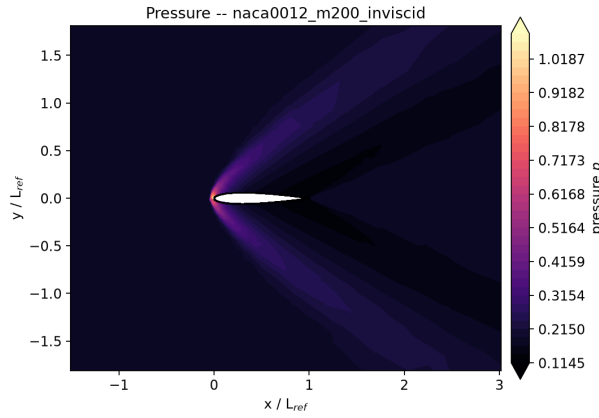


(a) Wall pressure coefficient: c_p , plotting C_p , from `surface.csv`. Wall pressure coefficient for `naca0012_m200_inviscid` versus chordwise station with the ordinate inverted; 404 surface samples, C_p range $[-0.354, 2.133]$.

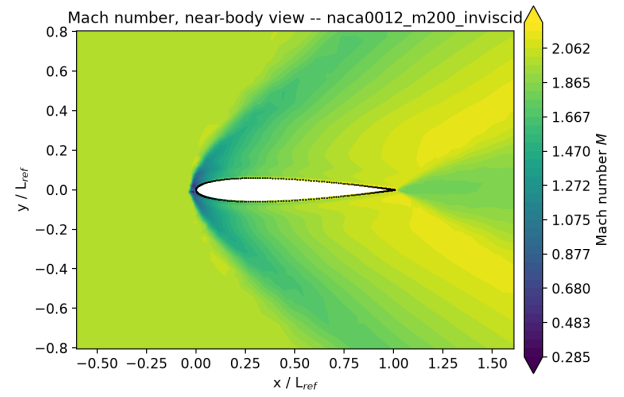


(b) Field contours, full window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.2854, 2.21]$ of the full range $[0.061, 2.722]$.

Figure 22: NACA0012, $M_\infty = 2.00$, inviscid: figure set 2 of 4.

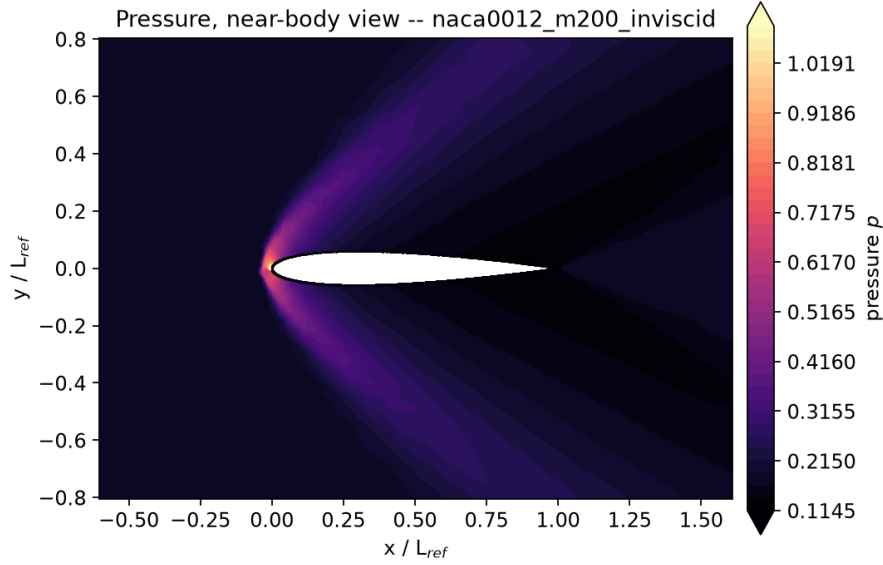


(a) Field contours, full window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.1145, 1.094]$ of the full range $[0.001985, 1.242]$.



(b) Field contours, near-body window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.2852, 2.21]$ of the full range $[0.061, 2.722]$.

Figure 23: NACA0012, $M_\infty = 2.00$, inviscid: figure set 3 of 4.



(a) Field contours, near-body window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile [0.1145, 1.094] of the full range [0.001985, 1.242].

Figure 24: NACA0012, $M_\infty = 2.00$, inviscid: figure set 4 of 4.

for each one.

A.5 NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$

Figures 29 to 32 reproduce every figure produced for `naca0012_m080_laminar_re5000` in the order they appear in `figure_manifest.csv`, which records the source data file and the plotted variable for each one.

A.6 NACA0012, $M_\infty = 2.00$, laminar $Re = 5000$

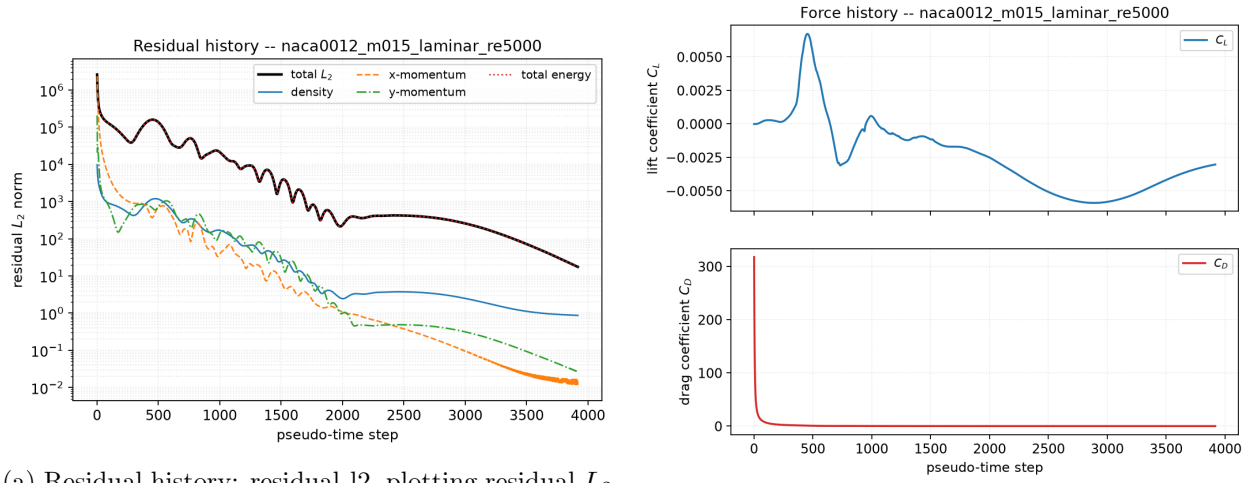
Figures 33 to 36 reproduce every figure produced for `naca0012_m200_laminar_re5000` in the order they appear in `figure_manifest.csv`, which records the source data file and the plotted variable for each one.

A.7 Cylinder, $M_\infty = 0.10$, laminar $Re = 20$

Figures 37 to 40 reproduce every figure produced for `cylinder_m010_laminar_re20` in the order they appear in `figure_manifest.csv`, which records the source data file and the plotted variable for each one.

A.8 Cylinder, $M_\infty = 0.10$, laminar $Re = 200$

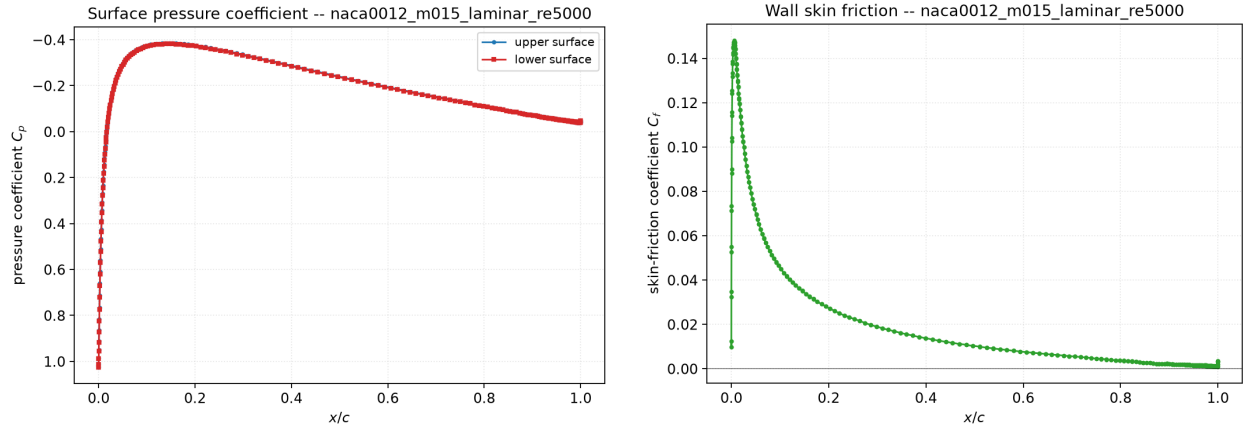
Figures 41 to 45 reproduce every figure produced for `cylinder_m010_laminar_re200` in the order they appear in `figure_manifest.csv`, which records the source data file and the plotted variable for each one.



(a) Residual history: residual_l2, plotting residual L_2 norm, from `residuals.csv`. Residual L_2 norm and the four conserved-component norms versus pseudo-time step for naca0012_m015_laminar_re5000; total reduction 5.17 orders of magnitude.

(b) Force history: cl_cd, plotting C_L and C_D , from `forces.csv`. Lift and drag coefficient history versus pseudo-time step for naca0012_m015_laminar_re5000; final $C_L = -0.00303332$, $C_D = 0.0563279$.

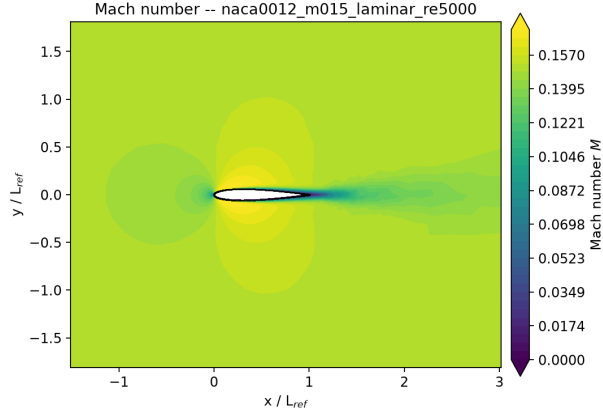
Figure 25: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$: figure set 1 of 4.



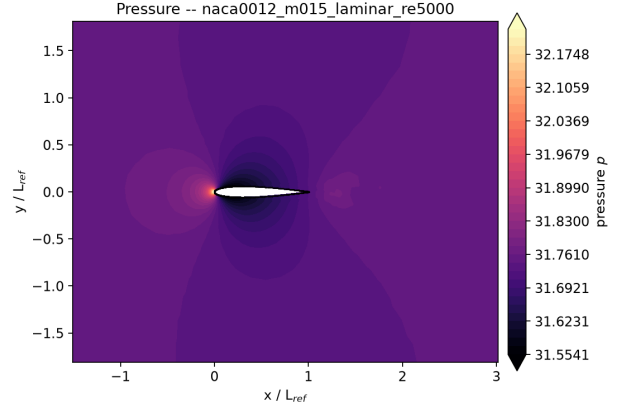
(a) Wall pressure coefficient: cp, plotting C_p , from `surface.csv`. Wall pressure coefficient for naca0012_m015_laminar_re5000 versus chordwise station with the ordinate inverted; 404 surface samples, C_p range $[-0.3859, 1.025]$.

(b) Wall skin-friction coefficient: cf, plotting C_f , from `surface.csv`. Wall skin-friction coefficient for naca0012_m015_laminar_re5000 versus chordwise station; C_f range $[0.000928, 0.1481]$ from tangential wall shear.

Figure 26: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$: figure set 2 of 4.

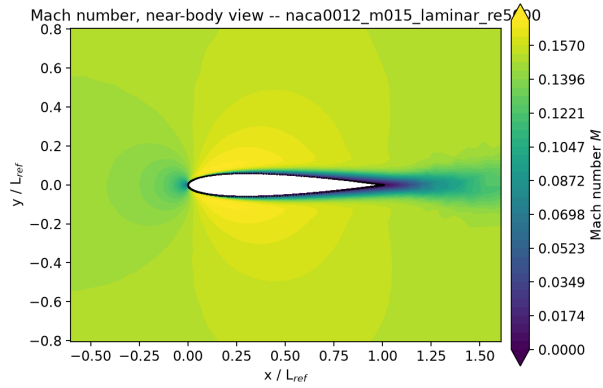


(a) Field contours, full window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[2.088\text{e-}06, 0.17]$ of the full range $[1.066\text{e-}06, 0.1726]$.

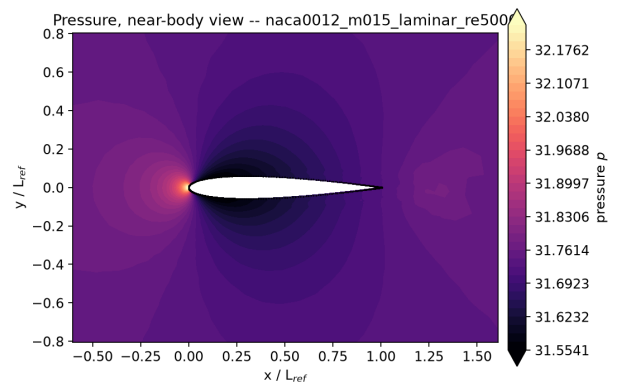


(b) Field contours, full window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[31.55, 32.23]$ of the full range $[31.55, 32.26]$.

Figure 27: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$: figure set 3 of 4.

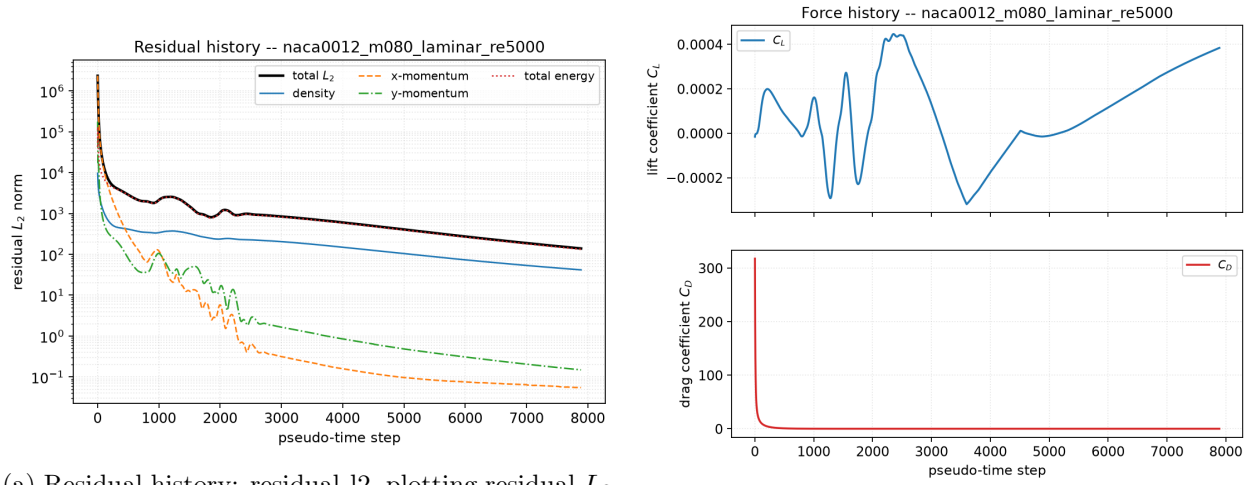


(a) Field contours, near-body window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[2.063\text{e-}06, 0.1701]$ of the full range $[1.066\text{e-}06, 0.1726]$.



(b) Field contours, near-body window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[31.55, 32.23]$ of the full range $[31.55, 32.26]$.

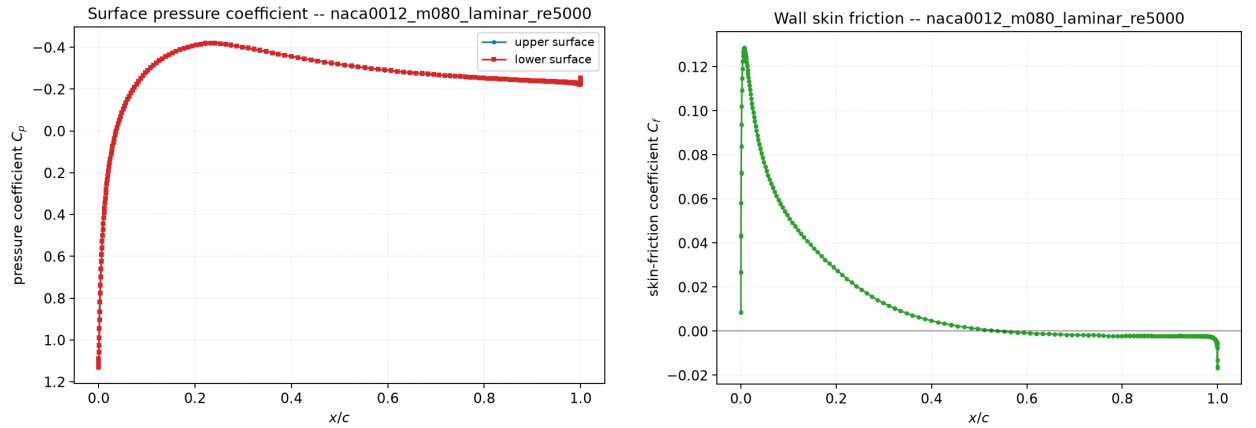
Figure 28: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$: figure set 4 of 4.



(a) Residual history: residual.l2, plotting residual L_2 norm, from `residuals.csv`. Residual L_2 norm and the four conserved-component norms versus pseudo-time step for naca0012_m080_laminar_re5000; total reduction 4.23 orders of magnitude.

(b) Force history: cl.cd, plotting C_L and C_D , from `forces.csv`. Lift and drag coefficient history versus pseudo-time step for naca0012_m080_laminar_re5000; final $C_L = 0.000384135$, $C_D = 0.0801972$.

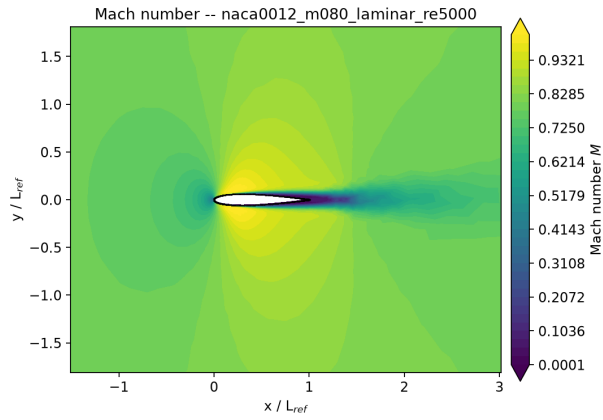
Figure 29: NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$: figure set 1 of 4.



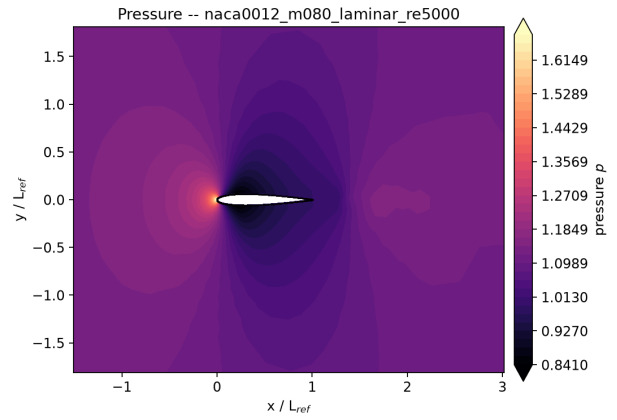
(a) Wall pressure coefficient: cp, plotting C_p , from `surface.csv`. Wall pressure coefficient for naca0012_m080_laminar_re5000 versus chordwise station with the ordinate inverted; 404 surface samples, C_p range [-0.4205, 1.13].

(b) Wall skin-friction coefficient: cf, plotting C_f , from `surface.csv`. Wall skin-friction coefficient for naca0012_m080_laminar_re5000 versus chordwise station; C_f range [-0.01669, 0.1287] from tangential wall shear.

Figure 30: NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$: figure set 2 of 4.

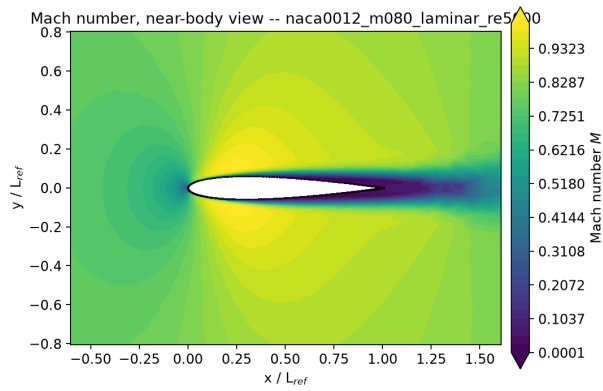


(a) Field contours, full window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[9.697\text{e-}05, 1.01]$ of the full range $[3.098\text{e-}05, 1.036]$.

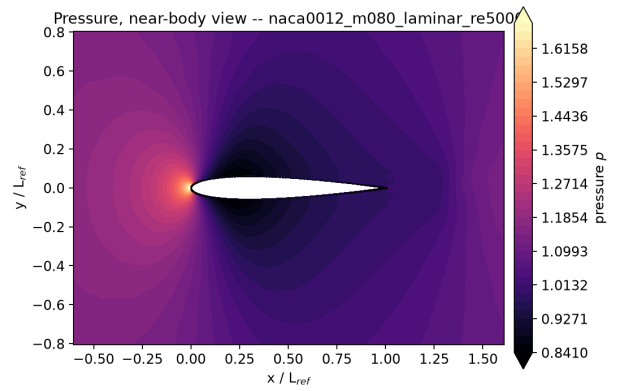


(b) Field contours, full window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.841, 1.679]$ of the full range $[0.8395, 1.71]$.

Figure 31: NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$: figure set 3 of 4.

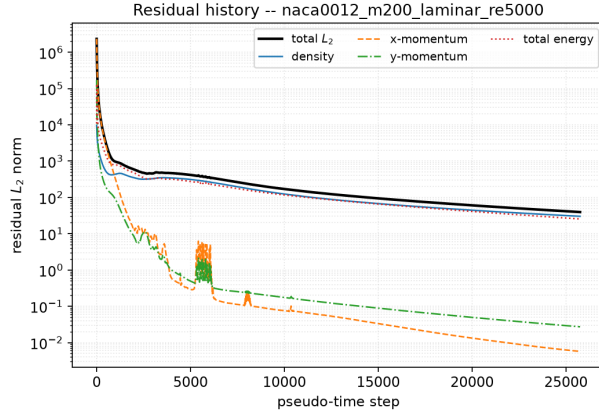


(a) Field contours, near-body window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[9.581\text{e-}05, 1.01]$ of the full range $[3.098\text{e-}05, 1.036]$.

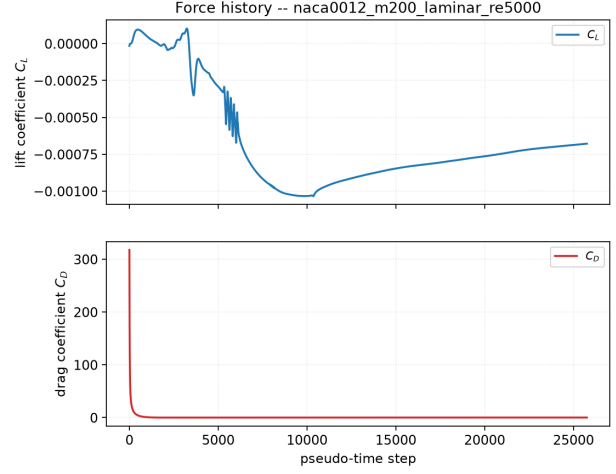


(b) Field contours, near-body window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.841, 1.68]$ of the full range $[0.8395, 1.71]$.

Figure 32: NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$: figure set 4 of 4.

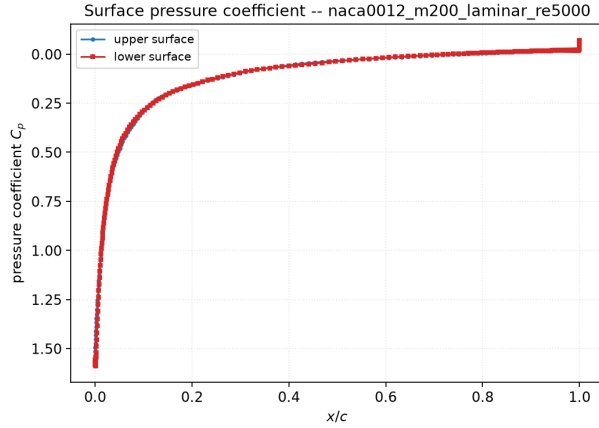


(a) Residual history: residual_l2, plotting residual L_2 norm, from `residuals.csv`. Residual L_2 norm and the four conserved-component norms versus pseudo-time step for naca0012_m200_laminar_re5000; total reduction 4.77 orders of magnitude.

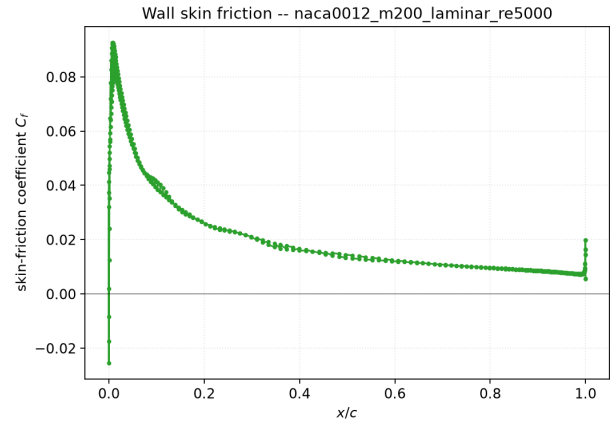


(b) Force history: cl_cd, plotting C_L and C_D , from `forces.csv`. Lift and drag coefficient history versus pseudo-time step for naca0012_m200_laminar_re5000; final $C_L = -0.000678145$, $C_D = 0.128008$.

Figure 33: NACA0012, $M_\infty = 2.00$, laminar $Re = 5000$: figure set 1 of 4.

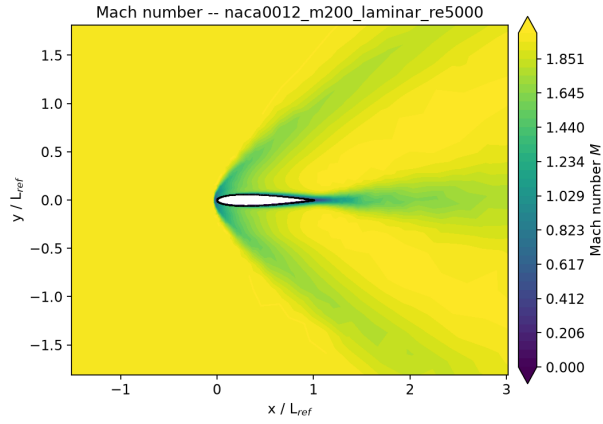


(a) Wall pressure coefficient: cp, plotting C_p , from `surface.csv`. Wall pressure coefficient for naca0012_m200_laminar_re5000 versus chordwise station with the ordinate inverted; 404 surface samples, C_p range $[-0.07117, 1.588]$.

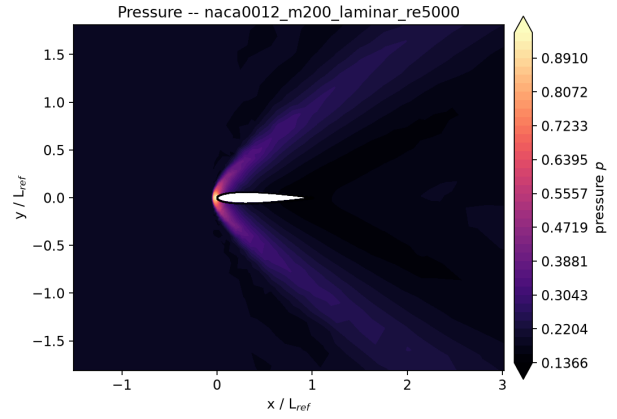


(b) Wall skin-friction coefficient: cf, plotting C_f , from `surface.csv`. Wall skin-friction coefficient for naca0012_m200_laminar_re5000 versus chordwise station; C_f range $[-0.02542, 0.09265]$ from tangential wall shear.

Figure 34: NACA0012, $M_\infty = 2.00$, laminar $Re = 5000$: figure set 2 of 4.

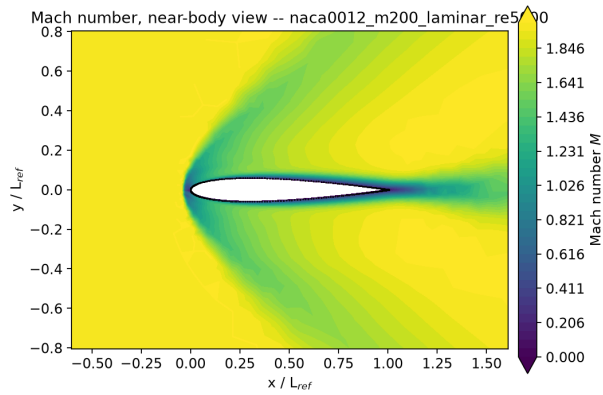


(a) Field contours, full window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.0004398, 2.005]$ of the full range $[5.108\text{e-}05, 2.087]$.

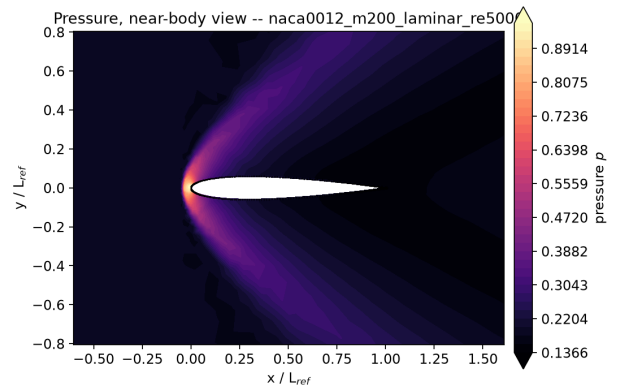


(b) Field contours, full window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.1366, 0.9538]$ of the full range $[0.1292, 0.9723]$.

Figure 35: NACA0012, $M_\infty = 2.00$, laminar $Re = 5000$: figure set 3 of 4.

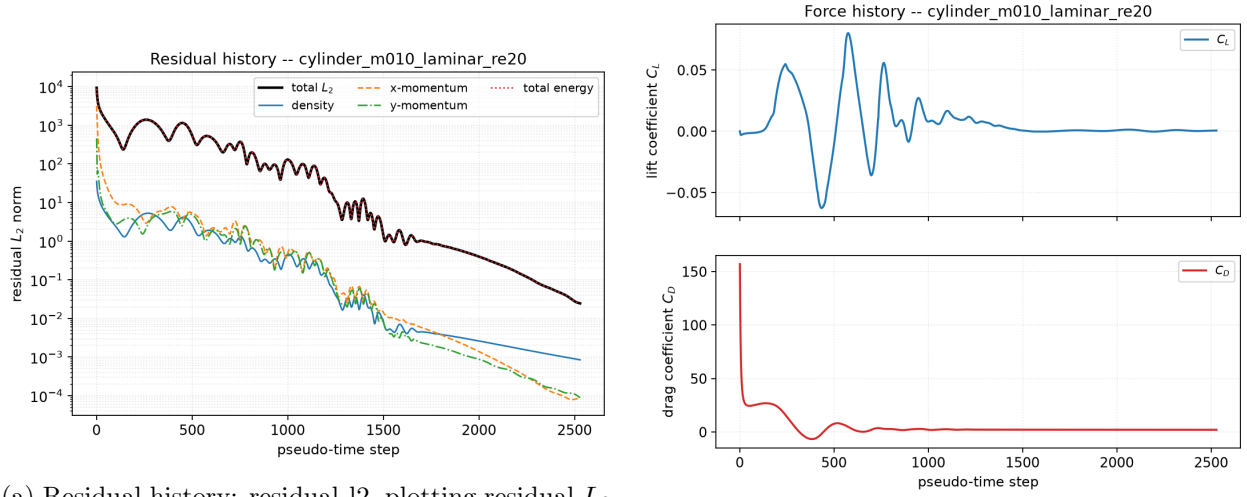


(a) Field contours, near-body window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.0004369, 2]$ of the full range $[5.108\text{e-}05, 2.084]$.



(b) Field contours, near-body window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.1366, 0.9543]$ of the full range $[0.1292, 0.9723]$.

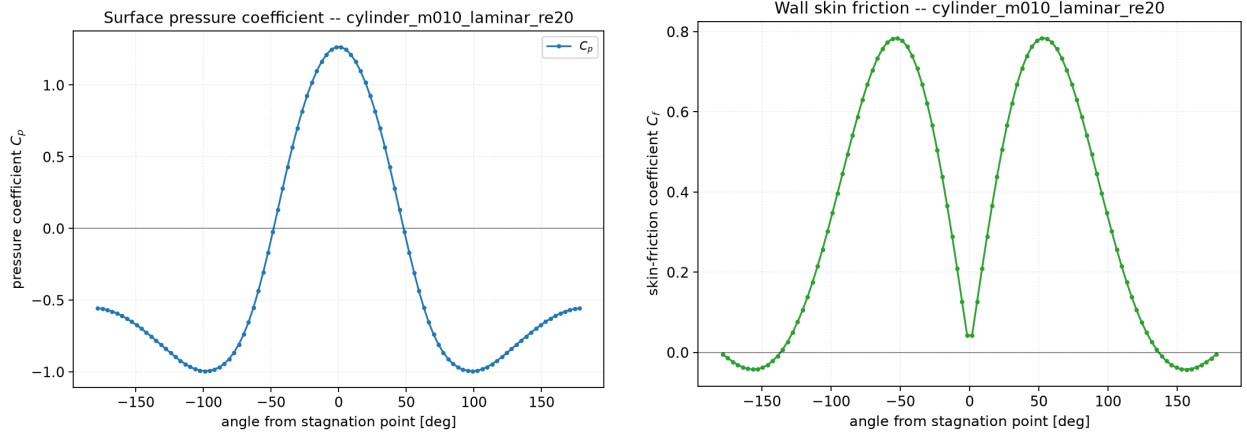
Figure 36: NACA0012, $M_\infty = 2.00$, laminar $Re = 5000$: figure set 4 of 4.



(a) Residual history: residual_l2, plotting residual L_2 norm, from `residuals.csv`. Residual L_2 norm and the four conserved-component norms versus pseudo-time step for cylinder_m010_laminar_re20; total reduction 5.58 orders of magnitude.

(b) Force history: cl_cd, plotting C_L and C_D , from `forces.csv`. Lift and drag coefficient history versus pseudo-time step for cylinder_m010_laminar_re20; final $C_L = 0.000614429$, $C_D = 2.01807$.

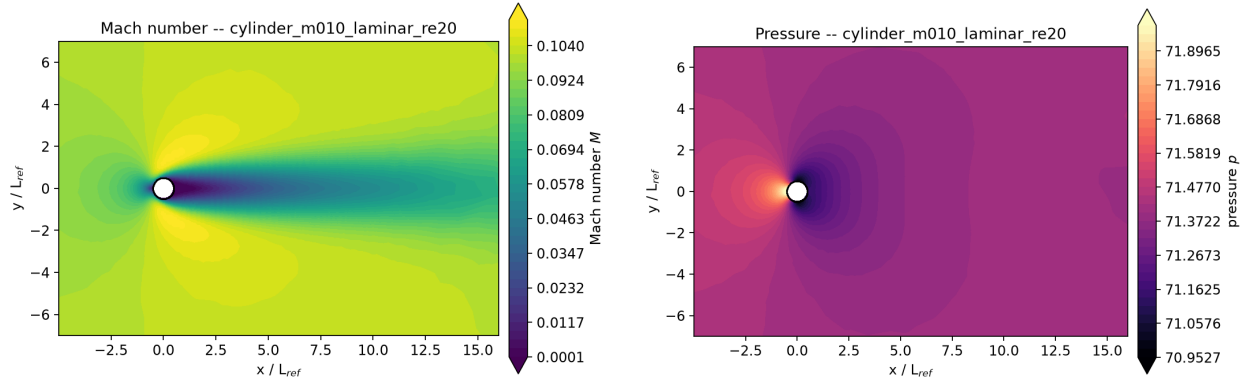
Figure 37: Cylinder, $M_\infty = 0.10$, laminar $Re = 20$: figure set 1 of 4.



(a) Wall pressure coefficient: cp, plotting C_p , from `surface.csv`. Wall pressure coefficient for cylinder_m010_laminar_re20 versus angle measured from the upstream stagnation point; 100 surface samples, C_p range $[-0.9949, 1.266]$.

(b) Wall skin-friction coefficient: cf, plotting C_f , from `surface.csv`. Wall skin-friction coefficient for cylinder_m010_laminar_re20 versus angle from the stagnation point; C_f range $[-0.04296, 0.784]$ from tangential wall shear.

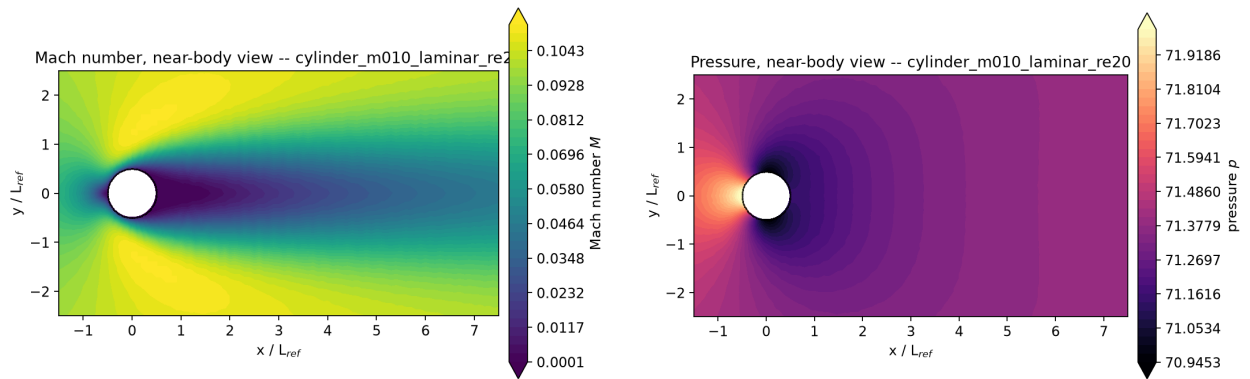
Figure 38: Cylinder, $M_\infty = 0.10$, laminar $Re = 20$: figure set 2 of 4.



(a) Field contours, full window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile [0.0001094, 0.1126] of the full range [5.107e-06, 0.1142].

(b) Field contours, full window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile [70.95, 71.98] of the full range [70.93, 72.06].

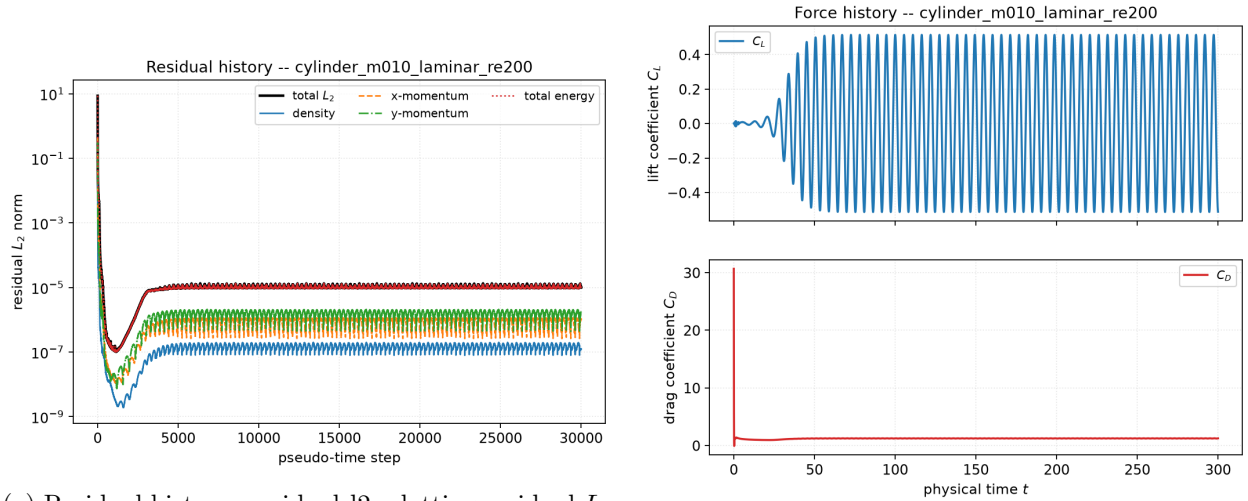
Figure 39: Cylinder, $M_\infty = 0.10$, laminar $Re = 20$: figure set 3 of 4.



(a) Field contours, near-body window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile [7.183e-05, 0.113] of the full range [5.107e-06, 0.1142].

(b) Field contours, near-body window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile [70.95, 72] of the full range [70.93, 72.06].

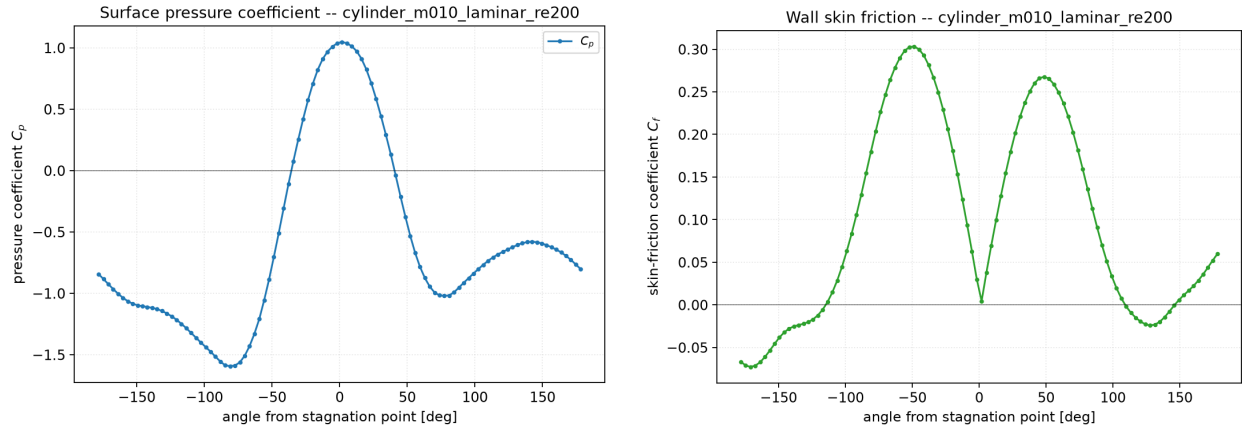
Figure 40: Cylinder, $M_\infty = 0.10$, laminar $Re = 20$: figure set 4 of 4.



(a) Residual history: residual.l2, plotting residual L_2 norm, from `residuals.csv`. Residual L_2 norm and the four conserved-component norms versus pseudo-time step for cylinder_m010_laminar_re200; total reduction 5.94 orders of magnitude.

(b) Force history: cl_cd, plotting C_L and C_D , from `forces.csv`. Lift and drag coefficient history versus physical time t for cylinder_m010_laminar_re200; final $C_L = -0.511083$, $C_D = 1.24829$.

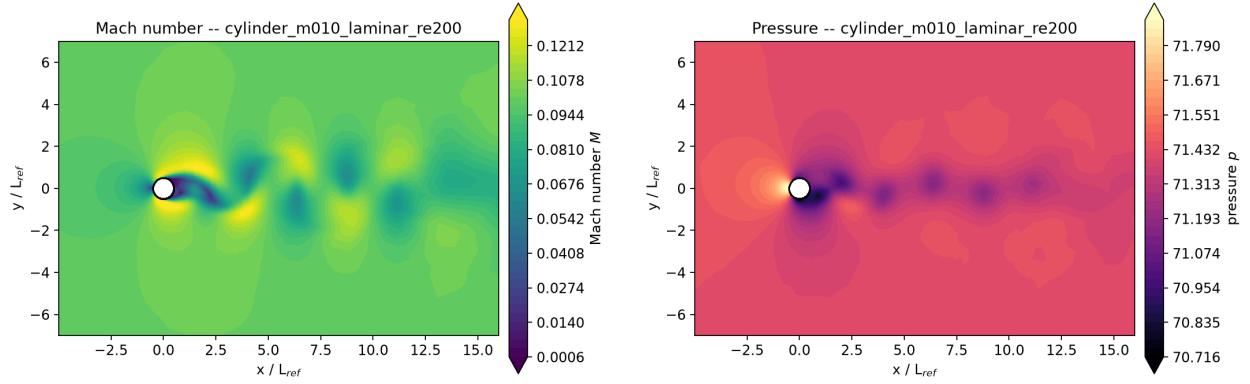
Figure 41: Cylinder, $M_\infty = 0.10$, laminar $Re = 200$: figure set 1 of 5.



(a) Wall pressure coefficient: cp, plotting C_p , from `surface.csv`. Wall pressure coefficient for cylinder_m010_laminar_re200 versus angle measured from the upstream stagnation point; 100 surface samples, C_p range [-1.593, 1.047].

(b) Wall skin-friction coefficient: cf, plotting C_f , from `surface.csv`. Wall skin-friction coefficient for cylinder_m010_laminar_re200 versus angle from the stagnation point; C_f range [-0.07283, 0.3032] from tangential wall shear.

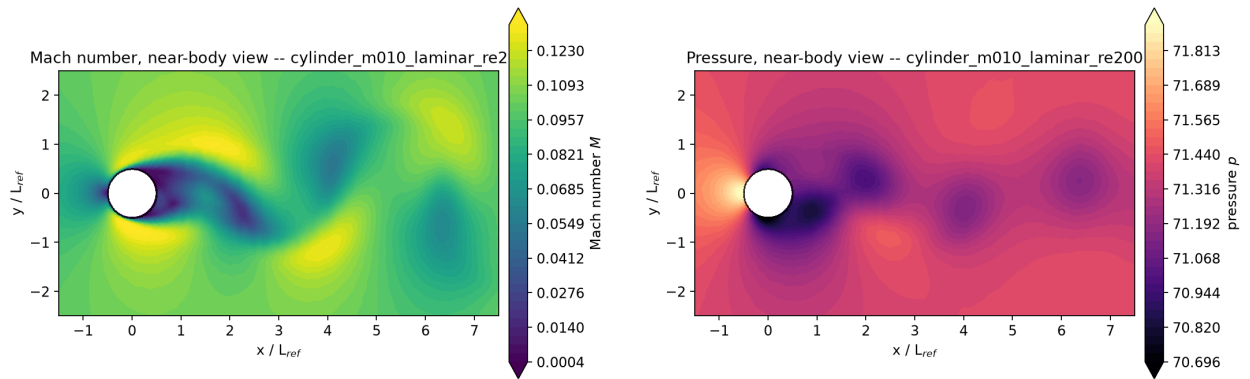
Figure 42: Cylinder, $M_\infty = 0.10$, laminar $Re = 200$: figure set 2 of 5.



(a) Field contours, full window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile [0.0005629, 0.1312] of the full range [3.065e-05, 0.1404].

(b) Field contours, full window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile [70.72, 71.88] of the full range [70.64, 71.95].

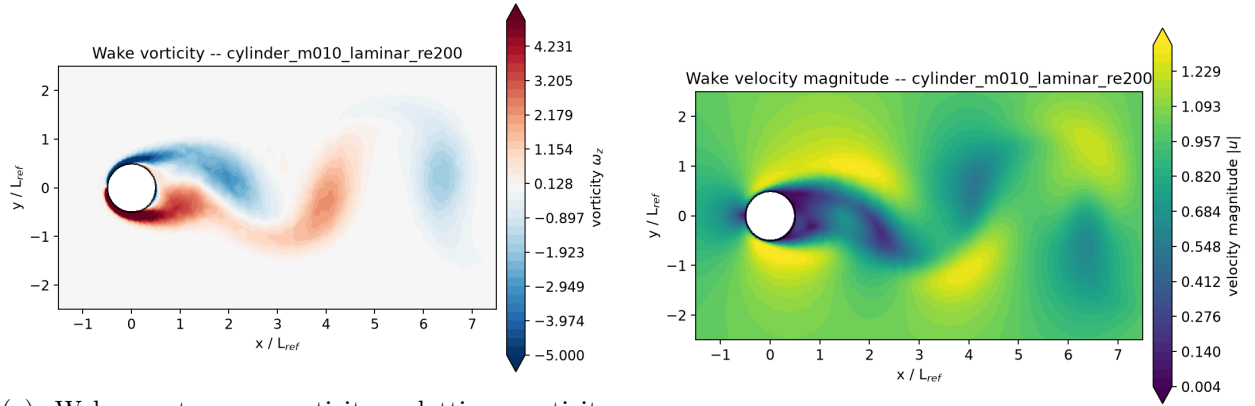
Figure 43: Cylinder, $M_\infty = 0.10$, laminar $Re = 200$: figure set 3 of 5.



(a) Field contours, near-body window: mach, plotting Mach number, from `field_final.vtu`. colour range clipped to the 1-99 percentile [0.0003942, 0.1332] of the full range [3.065e-05, 0.1404].

(b) Field contours, near-body window: pressure, plotting pressure, from `field_final.vtu`. colour range clipped to the 1-99 percentile [70.7, 71.91] of the full range [70.64, 71.95].

Figure 44: Cylinder, $M_\infty = 0.10$, laminar $Re = 200$: figure set 4 of 5.



(a) Wake contours: vorticity, plotting vorticity ω_z , from `field_final.vtu`. colour range clipped to the documented $[-5, 5]$ band from `NUMERICAL_PARAMETERS.md`; the raw field spans $[-26.92, 30.86]$.

(b) Wake contours: velocity_magnitude, plotting velocity magnitude $|\mathbf{u}|$, from `field_final.vtu`. colour range clipped to the 1-99 percentile $[0.003644, 1.331]$ of the full range $[4.725e-05, 1.402]$.

Figure 45: Cylinder, $M_\infty = 0.10$, laminar $Re = 200$: figure set 5 of 5.